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**(3GPP TS 37.579-5 version 18.5.0 Release 18)**

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In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

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y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

The present document is part 5 of a multi-part conformance test specification for Mission Critical services consisting of:

3GPP TS 37.579-1 [2]: "Mission Critical (MC) services; Part 1: Common test environment"

3GPP TS 37.579-2 [3]: "Mission Critical (MC) services; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification"

3GPP TS 37.579-4 [5]: "Mission Critical (MC) services; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification"

**3GPP TS 37.579-5: "Mission Critical (MC) services; Part 5: Abstract test suite (ATS)" (the present specification)**

3GPP TS 37.579-6 [59]: "Mission Critical (MC) services; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification"

3GPP TS 37.579-7 [60]: "Mission Critical (MC) services; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification"

In the present document, modal verbs have the following meanings:

**shall** indicates a mandatory requirement to do something

**shall not** indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

**should** indicates a recommendation to do something

**should not** indicates a recommendation not to do something

**may** indicates permission to do something

**need not** indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

**can** indicates that something is possible

**cannot** indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

**will** indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**will not** indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**might** indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

**might not** indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

**is** (or any other verb in the indicative mood) indicates a statement of fact

**is not** (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

# 1 Scope

The present document specifies the protocol and signalling conformance testing in TTCN-3 for the Mission Critical services defined by 3GPP.

The following TTCN test specification and design considerations can be found in the present document:

- the test system architecture;
- the overall test suite structure;
- the test models and ASP definitions;
- the test methods and usage of communication ports definitions;
- the test configurations;
- the design principles and assumptions;
- TTCN styles and conventions;
- the partial Implementation eXtra Information for Testing (IXIT) proforma;
- the test suites.

The Abstract Test Suites designed in the document are based on the test cases specified in 3GPP TS 37.579-2 [3], TS 37.579-6 [59] and TS 37.579-7 [60].

The applicability of the individual test cases is specified in the test ICS proforma specification in 3GPP TS 37.579-4 [5]). Where appropriate the Abstract Test Suites belonging to the present specification may refer to other Abstract Test Suites e.g. 3GPP TS 36.523-3 [27] and TS 38.523-3 [72] for test requirements related to the 3GPP bearers which carry the Mission Critical services data.

The present document is valid for TTCN development for Mission Critical services clients' conformance tests according to 3GPP Releases starting from Release 14 up to the Release indicated on the cover page of the present document.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
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[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 37.579-1: "Mission Critical (MC) services; Part 1: Common test environment".

[3] 3GPP TS 37.579-2: "Mission Critical (MC) services; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification".

[4] Void.

[5] 3GPP TS 37.579-4: "Mission Critical (MC) services; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification".

[6] 3GPP TS 36.523-1: "User Equipment (UE) conformance specification; Part 1: Protocol conformance specification"

[7] 3GPP TS 22.179: "Mission Critical Push To Talk (MCPTT) over LTE; Stage 1".

[8] Void.

[9] 3GPP TS 24.379: "Mission Critical Push To Talk (MCPTT) call control; Protocol specification".

[10] 3GPP TS 24.380: "Mission Critical Push To Talk (MCPTT) floor control; Protocol specification".

[11] 3GPP TS 24.481: "Mission Critical Services (MCS) group management; Protocol specification".

[12] 3GPP TS 24.482: "Mission Critical Services (MCS) identity management; Protocol specification".

[13] 3GPP TS 24.483: "Mission Critical Services (MCS) Management Object (MO)".

[14] 3GPP TS 24.484: "Mission Critical Services (MCS) configuration management; Protocol specification".

[15] 3GPP TS 33.179: "Security of Mission Critical Push-To-Talk (MCPTT)".

[16] 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

[17] 3GPP TS 24.237: "IP Multimedia Subsystem (IMS) Service Continuity; Stage 3".

[18] 3GPP TS 29.468: "Group Communication System Enablers for LTE (GCSE_LTE); MB2 Reference Point; Stage 3".

[19] 3GPP TS 24.301: "Non-Access-Stratum (NAS) protocol for Evolved Packet System (EPS); Stage 3".

[20] 3GPP TS 24.008: "Mobile Radio Interface Layer 3 specification; Core Network Protocols; Stage 3".

[21] 3GPP TS 23.003: "Numbering, addressing and identification".

[22] ISO/IEC 9646-1: "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 1: General concepts".

[23] ISO/IEC 9646-7: "Information technology - Open systems interconnection - Conformance testing methodology and framework - Part 7: Implementation Conformance Statements".

[24] 3GPP TS 23.303: "Proximity-based services (ProSe); Stage 2".

[25] IETF RFC 4566 (July 2006): "SDP: Session Description Protocol".

[26] 3GPP TS 26.171: "Speech codec speech processing functions; Adaptive Multi-Rate - Wideband (AMR-WB) speech codec; General description".

[27] 3GPP TS 36.523-3: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); User Equipment (UE) conformance specification; Part 3: Test suites".

[28] 3GPP TS 34.229-3: "Internet Protocol (IP) multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); User Equipment (UE) conformance specification; Part 3: Abstract Test Suites (ATS)".

[29] ISO/IEC 9646-1: "Information technology - Open Systems Interconnection - Conformance testing methodology and framework - Part 1: General concepts".

[30] ISO/IEC 9646-7: "Information technology - Open systems interconnection - Conformance testing methodology and framework - Part 7: Implementation Conformance Statements".

[31] ETSI ES 201 873-1: "Methods for Testing and Specification (MTS); The Testing and Test Control Notation version 3; Part 1: TTCN-3 Core Language".

[32] IETF RFC 3711: "The Secure Real-time Transport Protocol (SRTP)".

[33] 3GPP TS 27.007: "AT command set for User Equipment (UE)".

[34] IETF RFC 4661: "An Extensible Markup Language (XML)-Based Format for Event Notification Filtering".

[34] IETF RFC 4826: "Extensible Markup Language (XML) Formats for Representing Resource Lists".

[35] W3C: "XML Encryption Syntax and Processing Version 1.1", https://www.w3.org/TR/xmlenc-core1/.

[36] W3C: "XML Signature Syntax and Processing (Second Edition)", http://www.w3.org/TR/xmldsig-core/.

[37] Void.

[38] Void.

[39] Void.

[40] Void.

[41] W3C: "XML Encryption Syntax and Processing Version 1.1", https://www.w3.org/TR/xmlenc-core1/.

[42] W3C: "XML Signature Syntax and Processing (Second Edition)", http://www.w3.org/TR/xmldsig-core/.

[43] 3GPP TS 33.180: "Security of the mission critical service".

[44] IETF RFC 6507: "Elliptic Curve-Based Certificateless Signatures for Identity-Based Encryption (ECCSI)".

[45] IETF RFC 6508: "Sakai-Kasahara Key Encryption (SAKKE)".

[46] IETF RFC 6509 (February 2012): "MIKEY-SAKKE: Sakai-Kasahara Key Encryption in Multimedia Internet KEYing (MIKEY)".

[47] IETF RFC 3394: "Advanced Encryption Standard (AES) Key Wrap Algorithm".

[48] W3C: "XML Signature Syntax and Processing (Second Edition)", http://www.w3.org/TR/xmldsig-core/.

[49] IETF RFC 7515: "JSON Web Signature (JWS)".

[50] IETF RFC 5261: "An Extensible Markup Language (XML) Patch Operations Framework Utilizing XML Path Language (XPath) Selectors".

[51] IETF RFC 5874: "An Extensible Markup Language (XML) Document Format for Indicating a Change in XML Configuration Access Protocol (XCAP) Resources".

[52] IETF RFC 4354: "A Session Initiation Protocol (SIP) Event Package and Data Format for Various Settings in Support for the Push-to-Talk over Cellular (PoC) Service".

[53] IETF RFC 3986: "Uniform Resource Identifier (URI): Generic Syntax".

[54] 3GPP TS 23.280: "Common functional architecture to support mission critical services".

[55] 3GPP TS 24.281: "Mission Critical Video (MCVideo) signalling control; Protocol specification"

[56] 3GPP TS 24.581: "Mission Critical Video (MCVideo) media plane control; Protocol specification"

[57] 3GPP TS 24.282: "Mission Critical Data (MCData) signalling control; Protocol specification"

[58] 3GPP TS 24.582: "Mission Critical Data (MCData) media plane control; Protocol specification"

[59] 3GPP TS 37.579-6: "Mission Critical (MC) services; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification"

[60] 3GPP TS 37.579-7: "Mission Critical (MC) services; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification"

[61] 3GPP TS 36.508: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Common Test Environments for User Equipment (UE) Conformance Testing".

[62] IETF RFC 4122: "A Universally Unique IDentifier (UUID) URN Namespace"

[63] IETF RFC 5245: "Interactive Connectivity Establishment (ICE): A Protocol for Network Address Translator (NAT) Traversal for Offer/Answer Protocols"

[64] IETF RFC 3830: "MIKEY: Multimedia Internet KEYing".

[65] IETF RFC 6509 (February 2012): "MIKEY-SAKKE: Sakai-Kasahara Key Encryption in Multimedia Internet KEYing (MIKEY)".

[66] IETF RFC 6043: "MIKEY-TICKET: Ticket-Based Modes of Key Distribution in Multimedia Internet KEYing (MIKEY)".

[67] IETF RFC 3550: "RTP: A Transport Protocol for Real-Time Applications".

[68] IETF RFC 4975: "The Message Session Relay Protocol (MSRP)".

[69] IETF RFC 2616: "Hypertext Transfer Protocol -- HTTP/1.1".

[70] ETSI ES 201 873-11: "Methods for Testing and Specification (MTS); The Testing and Test Control Notation version 3; Part 11: Using JSON with TTCN-3".

[71] OMA-TS-REST_NetAPI_NMS-V1_0-20190528-C: "RESTful Network API for Network Message Storage".

[72] 3GPP TS 38.523-3: "5GS; User Equipment (UE) conformance specification; Part 3: Protocol Test Suites".

[73] 3GPP TS 38.508-1: "5GS; User Equipment (UE) conformance specification; Part 1: Common test environment".

[74] 3GPP TS 38.508-2: "5GS; User Equipment (UE) conformance specification; Part 2: Common Implementation Conformance Statement (ICS) proforma".

[75] ETSI ES 201 873-9: "Methods for Testing and Specification (MTS); The Testing and Test Control Notation version 3; Part 9: Using XML schema with TTCN-3".

[76] OMA-TS-REST_NetAPI_NotificationChannel-V1_0-20200319-C: “RESTful Network API for Notification Channel”.

# 3 Definitions of terms, symbols and abbreviations

## 3.1 Terms

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

In addition for the purposes of the present document, the following terms, definitions, symbols and abbreviations apply:

- such given in ISO/IEC 9646-1 [22] and ISO/IEC 9646-7 [23]

NOTE: Some terms and abbreviations defined in [22] and [23] are explicitly included below with small modification to reflect the terminology used in 3GPP.

**Implementation eXtra Information for Testing (IXIT)**: A statement made by a supplier or implementer of an UEUT which contains or references all of the information (in addition to that given in the ICS) related to the UEUT and its testing environment, which will enable the test laboratory to run an appropriate test suite against the UEUT.

**IXIT proforma:** A document, in the form of a questionnaire, which when completed for an UEUT becomes an IXIT.

**Protocol Implementation Conformance Statement (PICS):** An ICS for an implementation or system claimed to conform to a given protocol specification.

**Protocol Implementation eXtra Information for Testing (PIXIT):** An IXIT related to testing for conformance to a given protocol specification.

## 3.2 Symbols

No specific symbols have been identified so far.

## 3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

| | |
|---|---|
| ASP | Abstract Service Primitive |
| ICS | Implementation Conformance Statement |
| IXIT | Implementation eXtra Information for Testing |
| MC | Mission Critical |
| MCData | Mission Critical Data |
| MCPTT | Mission Critical Push To Talk |
| MCS | Mission Critical Services |
| MCVideo | Mission Critical Video |
| PTC | Parallel Test Component |
| RTCP | RTP Control Protocol |
| RTP | Real-time Transport Protocol |
| SRTCP | Secure RTCP |
| SRTP | Secure RTP |
| SS | System Simulator |
| SSRC | Synchronization SouRCe |
| TC | Test Case |
| UE | User Equipment |

# 4 Test system architecture

## 4.1 General system architecture

The architecture specified in TS 36.523-3 [27] clause 4.1.1 applies to the present document.

## 4.2 Component architecture

The architecture specified in TS 36.523-3 [27] clause 4.1.2 and TS 38.523-3 [72] clause 4.2 applies to the present document, with the clarification that only single RAT scenarios (E-UTRAN or NR/5GC) are considered by the test model of the present document.

# 5 Test models

## 5.1 MCX test model with TTCN based E-UTRA/EPC implementation (MCX EUTRA test model)

### 5.1.1 MCX Client on-network test model (EUTRA)

The MCX EUTRA on-network test model is depicted in figure 5.1.1-1. The test model consists of an IMS component and an HTTP component, on top of the multi-testers test model (E-UTRA) specified in TS 34.229-3 [28]. These parallel test components (PTCs) handle the IMS and HTTP signalling asynchronously.

The IMS PTC controls the IPCanEmu and the IP PTC. IPCanEmu is responsible for handling the E-UTRA cell(s) configuration in the SS as well as the E-UTRA/EPC level signalling and related procedures. The IPCanEmu and the NAS EMU are based on the TTCN implementation used for E-UTRA/EPC conformance testing according to TS 36.523-3 [27]. The IP PTC controls the IP related configurations as described in TS 36.523-3[27]. In addition, there is an SRTP port at the MCX IMS PTC mapped to the system interface to configure SRTP/SRTCP for media streams and media control messages (see clause 7.1.1.3). The media control messages are exchanged between TTCN and the system simulator via the IP PTC's IP_SOCK port (see clause 7.1.1.4 and clause 7.1.1.5) whereas the media stream messages are handled by the SS (see clause 7.1.1.5).

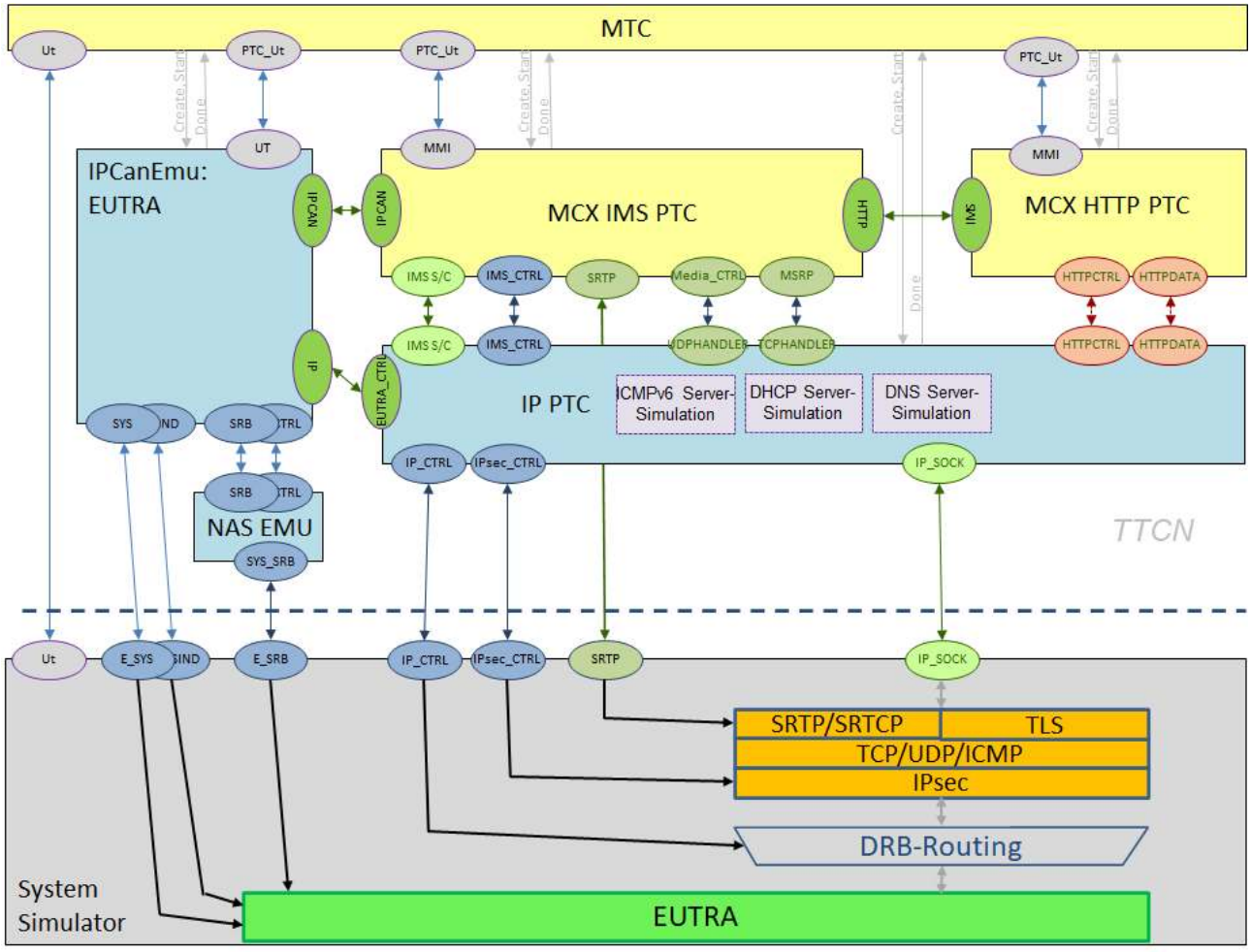



**Figure 5.1.1-1: MCX Client on-network test model with TTCN based E-UTRA/EPC implementation**

As described in TS 37.579-1 [2] clause 5.4.1A after switch on the UE may register to an internet PDN, an IMS PDN and the MCX PDN. Table 5.1.1-1 shows which IP addresses are assigned to a particular PDN.

**Table 5.1.1-1: Mapping of IP addresses to PDNs**

| PDN | IP address | Value | Comment |
|---|---|---|---|
| MCX | NW IP address | px_IPv4_Address1_NW, px_IPv6_Address1_NW | PDN_1 in TTCN |
| | UE IP address | px_IPv4_Address1_UE, px_IPv6_Address1_UE | |
| IMS | NW IP address | px_IPv4_Address2_NW, px_IPv6_Address2_NW | PDN_2 in TTCN |
| | UE IP address | px_IPv4_Address2_UE, px_IPv6_Address2_UE | |
| Internet | NW IP address | px_IPv4_Address3_NW, px_IPv6_Address3_NW | PDN_3 in TTCN |
| | UE IP address | px_IPv4_Address3_UE, px_IPv6_Address3_UE | |

### 5.1.2 MCX Client off-network test model (EUTRA)

An MCX Client off-network test model for EUTRA is not supported by the present version of the specification.

## 5.2 MCX test model with SS based E-UTRA/EPC or NR/5GC implementation (MCX IPCAN test model)

### 5.2.1 MCX Client on-network test model

In contrast to the MCX EUTRA test model (specified in clause 5.1.1) or NR5GC test model (specified in clause 5.3.1) the MCX IPCAN test model shown in figure 5.2.1-1 uses an external IPCAN (black-box) implementation which is integrated in the system simulator. In general, this IPCAN implementation shall be conformant in terms of TS 36.508 [61] and TS 38.508-1 [73] but this is out of the scope for the MCX IPCAN test model as long as the IPCAN implementation obeys the commands at the system interface as specified in annex E.

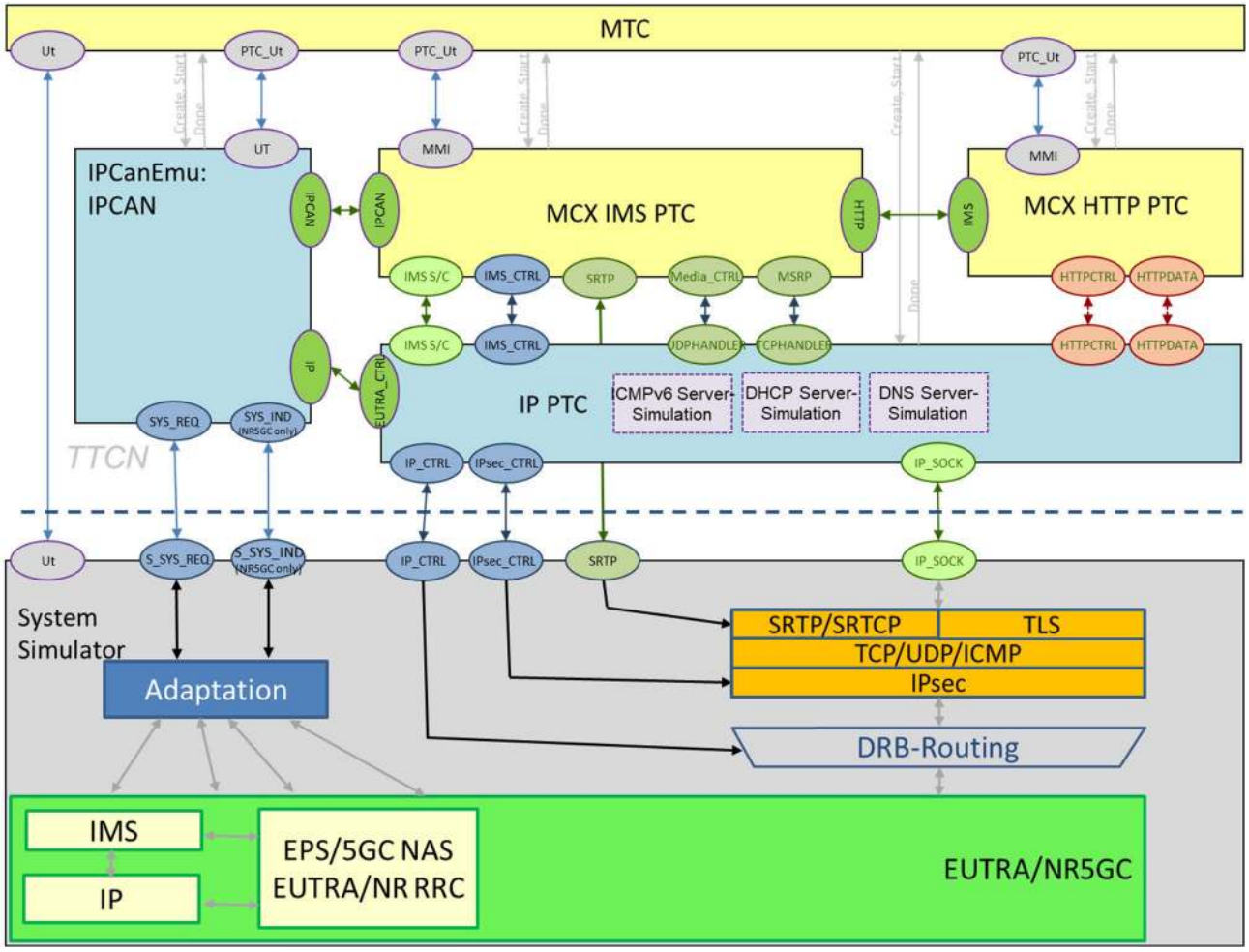



**Figure 5.2.1-1: MCX Client on-network test model with SS based E-UTRA/EPC or NR/5GC implementation**

NOTE 1: Whether or how the MCX IPCAN test model may use other IP-connectivity access networks is out of scope of this specification.

NOTE 2: The SYS_IND port shown in figure 5.2.1-1 is used only in case of NR5GC to indicate the PDU session id in a system indication and to assign the QFI in the corresponding acknowledgement.

NOTE 3: In the rest of this clause the term "PDN" is used for a PDN connectivity in EUTRA or a PDU session in NR5GC.

NOTE 4: The major procedures for configuration and control of the system simulator are illustrated in annex F.

As described in clauses 5.4.1A and 5.4.1B of TS 37.579-1 [2] after switch on the UE may register to an internet PDN, an IMS PDN and the MCX PDN. Nevertheless, in TTCN the MCX IPCAN test model supports signalling for the MCX PDN only: The IPCAN test model does not handle any signalling for the other PDNs and especially not the SIP signalling for an IMS registration to the IMS PDN.
⇒ It is up to SS implementation to handle the potential registrations to an internet PDN and/or an IMS PDN.

Tables 5.2.1-1 and 5.2.1-2 show the information provided to the SS for configuration of the PDNs for EUTRA and NR5GC.

**Table 5.2.1-1: PDN info for EUTRA**

| PDN | Parameter | Value/Comment |
|---|---|---|
| MCX | EPS Bearer ID (NOTE 2) | px_MCX_IPCAN_EpsBearerId_MCX (NOTE 1) |
| | NW IP address | px_IPv4_Address1_NW, px_IPv6_Address1_NW (NOTE 3) |
| | UE IP address | px_IPv4_Address1_UE, px_IPv6_Address1_UE (NOTE 3) |
| IMS | EPS Bearer ID (NOTE 2) | px_MCX_IPCAN_EpsBearerId_IMS (NOTE 1) |
| | NW IP address | px_IPv4_Address2_NW, px_IPv6_Address2_NW |
| | UE IP address | px_IPv4_Address2_UE, px_IPv6_Address2_UE |
| Internet | EPS Bearer ID (NOTE 2) | px_MCX_IPCAN_EpsBearerId_INTERNET (NOTE 1) |
| | NW IP address | px_IPv4_Address3_NW, px_IPv6_Address3_NW |
| | UE IP address | px_IPv4_Address3_UE, px_IPv6_Address3_UE |
| NOTE 1: Values depending on SS implementation<br>NOTE 2: The DRB associated with the EPS bearer shall have the DRB ID := EPS Bearer ID – 4.<br>NOTE 3: Even though for MCX the user plane signalling is handled in TTCN still the SS needs the IP addresses to be provided to the UE in NAS signalling. | | |

**Table 5.2.1-2: PDU session info for NR5GC**

| PDN | Parameter | Value/Comment |
|---|---|---|
| MCX | APN | pc_APN_ID_MCX (NOTE 2) |
| | NW IP address | px_IPv4_Address1_NW, px_IPv6_Address1_NW (NOTE 3) |
| | UE IP address | px_IPv4_Address1_UE, px_IPv6_Address1_UE (NOTE 3) |
| | QFI (NOTE 1) | 11 (NOTE 4) |
| IMS | APN | pc_APN_ID_IMS (NOTE 2) |
| | NW IP address | px_IPv4_Address2_NW, px_IPv6_Address2_NW |
| | UE IP address | px_IPv4_Address2_UE, px_IPv6_Address2_UE |
| Internet | APN | pc_APN_ID_Internet (NOTE 2) |
| | NW IP address | px_IPv4_Address3_NW, px_IPv6_Address3_NW |
| | UE IP address | px_IPv4_Address3_UE, px_IPv6_Address3_UE |
| NOTE 1: Provided in acknowledgement of system indication indicating the PDU session id.<br>NOTE 2: Defined in TS 38.508-2 [74] Table A.4.4-2A.<br>NOTE 3: Even though for MCX the user plane signalling is handled in TTCN still the SS needs the IP addresses to be provided to the UE in NAS signalling.<br>NOTE 4: According to TS 38.508-1 [73] Table 4.8.2.1-10 and Table 4.8.2.3-9. | | |

Dedicated EPS bearers and QoS flows use packet filters according to TS 36.508 [61] clause 6.6.2 and TS 38.508-1 [73] clause 4.8.2.2 as shown in table 5.2.1-3.

**Table 5.2.1-3: Packet filters for dedicated EPS bearers and QoS flows**

| MC Service | Value/Comment | | |
|---|---|---|---|
| MCPTT | Audio | Remote port range | tsc_MCX_MediaPort_Audio .. tsc_MCX_MediaPort_Audio + 1 |
| | | Protocol | UDP |
| | Media Control | Single remote port | tsc_MCX_MediaControlPortNW |
| | | Protocol | UDP |
| MCVideo | Audio | Remote port range | tsc_MCX_MediaPort_Audio .. tsc_MCX_MediaPort_Audio + 1 |
| | | Protocol | UDP |
| | Video | Remote port range | tsc_MCX_MediaPort_Video .. tsc_MCX_MediaPort_Video + 1 |
| | | Protocol | UDP |
| | Media Control | Single remote port | tsc_MCX_MediaControlPortNW |
| | | Protocol | UDP |
| MCData | | Single remote port | tsc_MCData_MediaPort_MSRP |
| | | Protocol | TCP |

The respective packet filter is provided to the SS together with the EPS bearer id (EUTRA) or the QoS flow (NR5GC) at call establishment. EPS bearer ids and QoS flows are as shown in table 5.2.1-4.

**Table 5.2.1-4: Dedicated EPS bearer ids and QoS flows**

| MC Service | EPS bearer id (EUTRA; NOTE 1) | QoS flow (NR5GC) | |
|---|---|---|---|
| | | PDU session id | QFI |
| MCPTT | px_MCX_IPCAN_DedicatedEpsBearerId_MCPTT (NOTE 2) | (NOTE 3) | 12 (NOTE 4) |
| MCVideo | px_MCX_IPCAN_DedicatedEpsBearerId_MCVideo (NOTE 2) | (NOTE 3) | 13 (NOTE 4) |
| MCData | px_MCX_IPCAN_DedicatedEpsBearerId_MCData (NOTE 2) | (NOTE 3) | 14 (NOTE 4) |
| NOTE 1: The DRB associated with the EPS bearer shall have the DRB ID := EPS Bearer ID – 4.<br>NOTE 2: Values depending on SS implementation.<br>NOTE 3: As chosen by the UE for the MCX PDU session at registration.<br>NOTE 4: According to TS 38.508-1 [73] Table 4.8.2.1-11 and Table 4.8.2.3-10. | | | |

# 5.3 MCX test model with TTCN based NR/5GC implementation (MCX NR5GC test model)

## 5.3.1 MCX Client on-network test model (NR5GC)

The MCX NR5GC on-network test model is depicted in figure 5.3.1-1.

The MCX NR5GC on-network test model is basically the same as described in clause 5.1.1 for E-UTRA with the following clarifications:

- In TTCN the NR5GC's IPCanEmu PTC and the NAS EMU NR PTCs replace the EUTRA's IPCanEmu and NAS EMU PTCs.
- The IPCanEmu and the NAS EMU NR are based on the TTCN implementation used for NR/5GC conformance testing according to TS 38.523-3 [72].
- At the system simulator an NR protocol stack is used instead of the E-UTRA protocol stack.

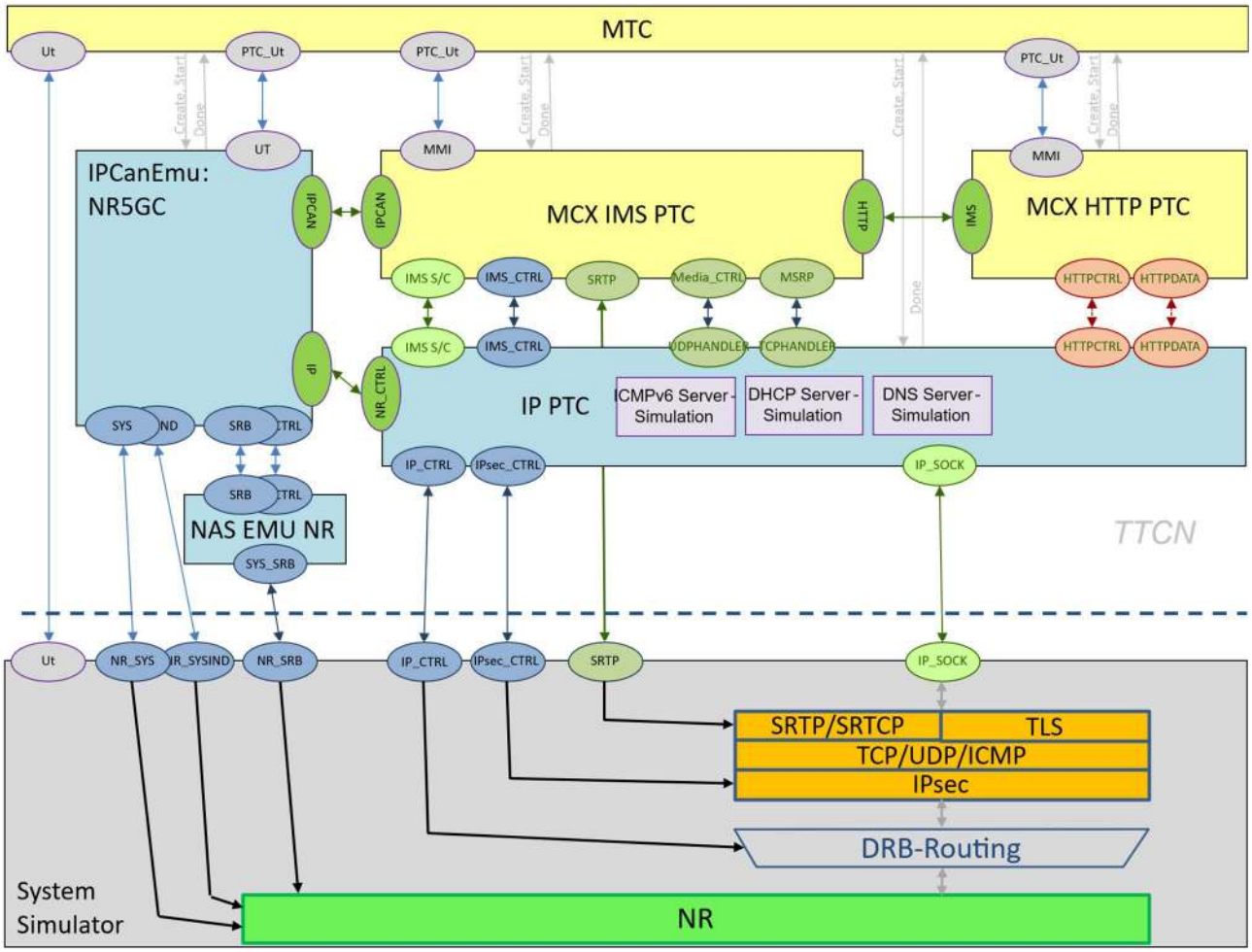



**Figure 5.3.1-1: MCX Client on-network test model with TTCN based NR/5GC implementation**

As described in TS 37.579-1 [2] clause 5.4.1B after switch on the UE may request an internet PDU session, an IMS PDU session and the MCX PDU session. Table 5.3.1-1 shows which IP addresses are assigned to a particular PDU session.

**Table 5.3.1-1: Mapping of IP addresses to PDU sessions**

| PDU session | IP address | Value | Comment |
|---|---|---|---|
| MCX | NW IP address | px_IPv4_Address1_NW, px_IPv6_Address1_NW | PDN_1 in TTCN |
| | UE IP address | px_IPv4_Address1_UE, px_IPv6_Address1_UE | |
| IMS | NW IP address | px_IPv4_Address2_NW, px_IPv6_Address2_NW | PDN_2 in TTCN |
| | UE IP address | px_IPv4_Address2_UE, px_IPv6_Address2_UE | |
| Internet | NW IP address | px_IPv4_Address3_NW, px_IPv6_Address3_NW | PDN_3 in TTCN |
| | UE IP address | px_IPv4_Address3_UE, px_IPv6_Address3_UE | |

## 5.3.2 MCX Client off-network test model (NR5GC)

An MCX Client off-network test model for NR/5GC is not supported by the present version of the specification.

# 6 System interface

## 6.1 Upper tester interface

The Upper Tester (UT) interface is the same as defined in TS 36.523-3 [27] clause 5. In addition there are the MCX specific MMI commands of table 6.1-1 and the MCX specific AT commands of table 6.1-2.

**Table 6.1-1: MMI commands**

| Command | Parameters | | ResultString |
|---|---|---|---|
| | Name | Value | |
| “MCX_USERACTION” | “UserAction” | <charstring> | (none) |
| “MCX_USERCHECK” | “UserCheck” | <charstring> | "YES" if the check is successful, "NO" otherwise |

**Table 6.1-2.: AT Commands**

| Command | Reference |
|---|---|
| AT+CAPTT | TS 27.007 [33] |

## 6.2 Abstract system primitives

This clause specifies the abstract system primitives (ASPs) used on the system interface to configure and control the SS. The MCX test system interface re-uses the ASPs specified in TS 36.523-3 [27] (see clause 6 and Annex D), TS 38.523-3 [72] (see clause 6.2 and Annex D) and TS 34.229-3 [28] (see clause 6).

# 7 Test methods and design considerations

## 7.0 Introduction

The present document focusses on on-network client test cases for MCPTT (TS 37.579-2 [3]), MCVideo (TS 37.579-6 [59]) and MCData (TS 37.579-7 [60]). The major test behaviour and the verdict assignment of the test cases happens at the MCX IMS PTC and the MCX HTTP PTC as shown in the figures of clause 5.

## 7.1 Media plane signalling

### 7.1.0 Introduction

In addition to the HTTP and SIP signalling over control plane as used for the procedures in TS 24.379 [9], TS 24.281 [55] and TS 24.281 [57] there are MCX-specific procedures using the media plane as specified in TS 24.380 [10], TS 24.581 [56] and TS 24.581 [58].

### 7.1.1 MCPTT

#### 7.1.1.1 Void

#### 7.1.1.2 UDP/IP handling

The mechanisms specified in TS 36.523-3 [27] clause 4.2.4 are applied for UDP/IP configuration and Routing Table configuration.

#### 7.1.1.3 RTP/RTCP handling

The RTP/RTCP loopback mechanism specified in TS 36.523-3 [27] clause 4.2.4.4 applies as baseline.

As MCPTT test cases require SRTP (secure RTP) for the media stream, the loopback mechanism needs to be enhanced: An RTP/RTCP packet received in uplink needs to be decrypted with the Rx or uplink key, encrypted with the Tx or downlink key and then sent back to the UE.

TTCN controls the usage of security for SRTP/SRTCP and provides the necessary security parameters to the SS via the SRTP port. Once security has been configured by TTCN, the SS shall handle media plane encryption and decryption.

The SRTP system interface is defined in Annex C.

#### 7.1.1.4 Media Control handling

MCPTT requires that media control messages for floor control and pre-established session control according to TS 24.380 [10] are made available and handled in TTCN as structured messages.

TTCN uses the IP PTC to handle the UDP socket for media plane control messages: UDP packet information consisting of the encoded UDP payload and the local and remote socket information (IP address and port) is exchanged between the MCX IMS PTC and the IP PTC. The Client's port number is retrieved from the SDP signalling during connection establishment.

Media plane security is configured at the SRTP system port: The SS gets the information how to identify the UDP packets to be encrypted and decrypted together with the key material and the security policy parameters (see annex C).

#### 7.1.1.5 SS pseudo-algorithm for RTP / RTCP / Media Plane Control handling

The following summarizes the requirements on the SS with regards to RTP / RTCP / Media Control handling within the SS:

- Uplink direction:
  - Upon reception of an SRTP/SRTCP packet, the SS shall extract the UDP payload and decrypt it using the RX crypto parameters of the respective connection.
  - In case of an RTP packet or an RTCP packet associated with the (RTP) media stream the SS shall apply the RTP/RTCP loopback as specified in clause 4.2.4.4 of TS 36.523-3 [27] with the following clarification:
    The SS shall replace the SSRC in the SRTP/SRTCP packet (octets 8 to 11 for SRTP and octets 4 to 7 for SRTCP, see IETF RFC 3711 [32] clauses 3.1 and 3.4) by the SSRC as configured at the SS for this connection.
    The SS shall encrypt the packet using the crypto parameters configured for the connection.
  - For RTCP APP packets the generic mechanisms of the IP test model according to clause 4.2 of TS 36.523-3 [27] apply: A UDP socket is configured by TTCN with corresponding entry in the common IP routing table so that the UDP packets are routed to the IP PTC which routes them to the MCX IMS PTC via the UDPHANDLER/Media_CTRL ports. Decoding of the media control messages is done in TTCN using TTCN-3's decvalue function.
- Downlink direction:
  - TTCN encodes media control messages using TTCN-3's encvalue function and sends the resulting datagram via the UDPHANDLER/Media_CTRL ports to the IP PTC and from there to the UDP socket configured at the SS. The SS shall encrypt the packet as according to the SRTP configuration for the connection and send it to the remote end.

### 7.1.2 MCVideo

The same principles apply as specified for MCPTT in clause 7.1.1 with the difference that the media plane control signalling is the transmission control protocol of TS 24.581 [56].

### 7.1.3 MCData

For MCData the Message Session Relay Protocol (MSRP) according to RFC 4975 [68] is used for test cases using the media plane. MSRP uses TCP as transport protocol.

The encoding, decoding and handling of the MSRP messages is done at the MCX IMS PTC which exchanges the encoded MSRP messages with the generic TCP handler of the IP PTC as shown the figures of clause 5.

The routing of the MSRP IP data is configured according to the IP Test Model of TS 36.523-3 [27] clause 4.2.4.

## 7.2 MCX functional architecture: HTTP, SIP

### 7.2.1 HTTP-Signalling

Figure 7.2.1-1 shows the functional architecture for HTTP-Signalling between the MCX-capable UE and different network entities as emulated by the test models.

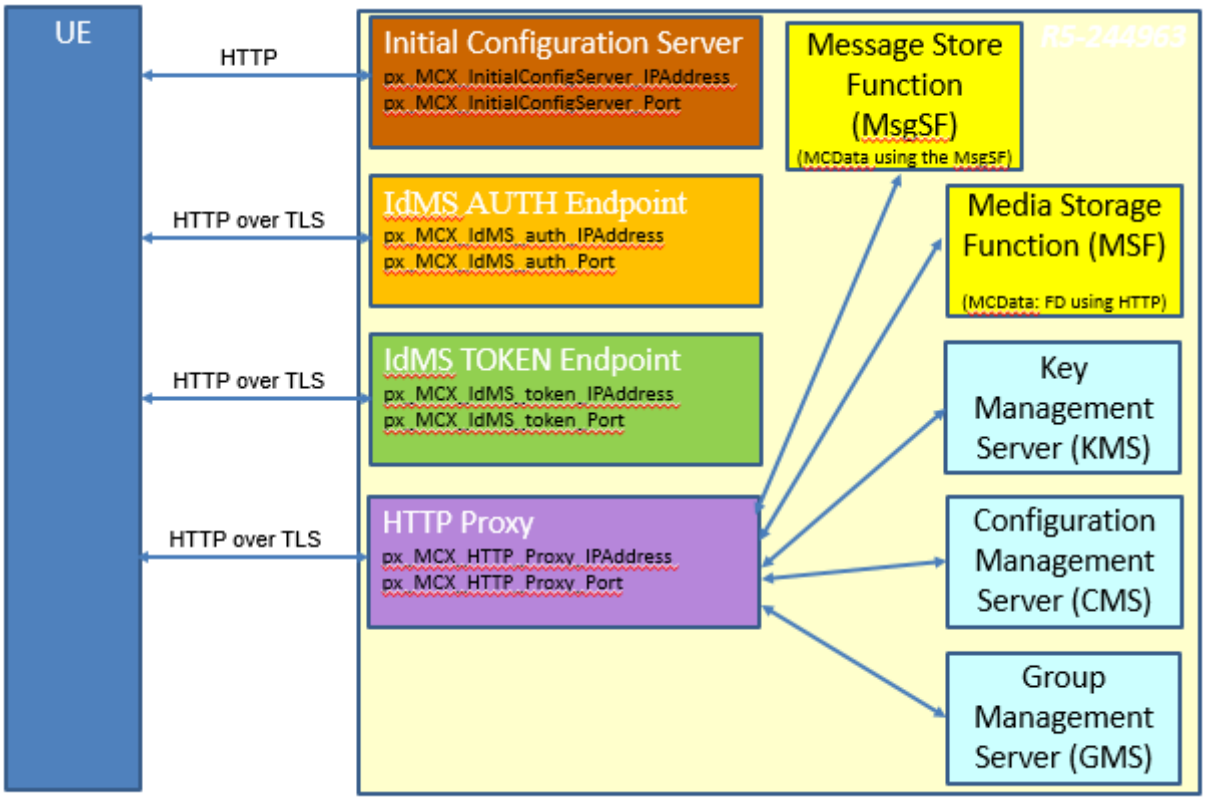



**Figure 7.2.1-1: Functional architecture for HTTP-Signalling**

The UE is expected to establish HTTP connections to the following entities:

- Initial Configuration Server:
  The UE gets configured with IP address (px_MCX_InitialConfigServer_IPAddress) and port number (px_MCX_InitialConfigServer_Port) of the server from which it retrieves the MCS UE initial configuration document (TS 24.484 [14] clause 7.2).
  It depends on UE implementation how the IP address and port are configured at the UE but the UE is expected to support this to be configured.
  As there is no requirement by the core specs the HTTP connection is not security protected.

- IdMS AUTH Endpoint:
  The UE retrieves IP address (px_MCX_IdMS_auth_IPAddress) and port number (px_MCX_IdMS_auth_Port)

from the MCS UE initial configuration document (TS 24.484 [14] clause 7.2).
The connection shall be TLS protected.

- IdMS TOKEN Endpoint:
  The UE retrieves IP address (px_MCX_IdMS_token_IPAddress) and port number (px_MCX_IdMS_token_Port) from the MCS UE initial configuration document (TS 24.484 [14] clause 7.2).
  The connection shall be TLS protected.

- HTTP Proxy:
  The UE retrieves IP address (px_MCX_HTTP_Proxy_IPAddress) and port number (px_MCX_HTTP_Proxy_Port) from the MCS UE initial configuration document (TS 24.484 [14] clause 7.2).
  The connection shall be TLS protected.

Design considerations and working assumptions:

- Domain Name Server
  DNS is not needed for address resolution as in the URIs of the network entities in the MCS UE initial configuration document the IP address of the respective entity can be used as host name.

- Initial UE configuration
  The UE is able to retrieve the MCS UE initial configuration document (TS 24.484 [14] clause 7.2) via an HTTP connection to an Initial Configuration Server. Other means of initial configuration are out of scope for conformance testing.

- HTTP signalling between UE and KMS
  The UE is expected to support HTTP signalling with the KMS via the HTTP proxy using a transport key (TrK) according to TS 33.180 [43] clause 4.3.3 and clause 5.3.2.

- Use of TLS
  All HTTP traffic between the UE and the HTTP proxy and between the UE and the IdMS is protected using HTTPS (TS 33.180 [43] clause 5.1.3.1, clause 6.1.1 and B.12).
  NOTE: Even though NOTE 2 of TS 24.482 [12] clause 6.2.1 might be interpreted as if the HTTP signalling to the authorization endpoint may not be TLS secured, 33.180 [43] mandates TLS to be used.

- IP Addresses and port numbers
  IP addresses and port numbers of the network entities as shown in figure 7.2.1-1 are given as PIXIT (see clause 9). As only restriction the tuple {IP address, port number} needs to be unique for each entity, but there is no restriction no to use one and the same IP address for all entities as long as they use different port numbers.

- According to TS 24.282 [57] clause 10.2.2.1 the HTTP signalling between the MCData client and the media storage function (MSF) happens via an HTTP proxy according to annex A of TS 24.482 [12]. It is assumed that this is the same proxy as used for signalling with the KMS, CMS and GMS.

- According to TS 24.282 [57] clause 21.1 the HTTP signalling between the MCData message store client and MCData message store function (MsgSF) happens via an HTTP proxy according to annex A of TS 24.482 [12]. It is assumed that this is the same proxy as used for signalling with the KMS, CMS and GMS.

- According to RFC 2616 [69] clause 5.1.2 HTTP requests via an HTTP proxy require the Request-URI to be an absolute URI ⇒ HTTP requests to KMS, CMS and GMS shall have an absolute URI as Request-URI.

## 7.2.2 SIP-Signalling

Figure 7.2.2-1 shows the functional architecture for SIP-Signalling between the MCX-capable UE and different network entities as emulated by the test model(s).

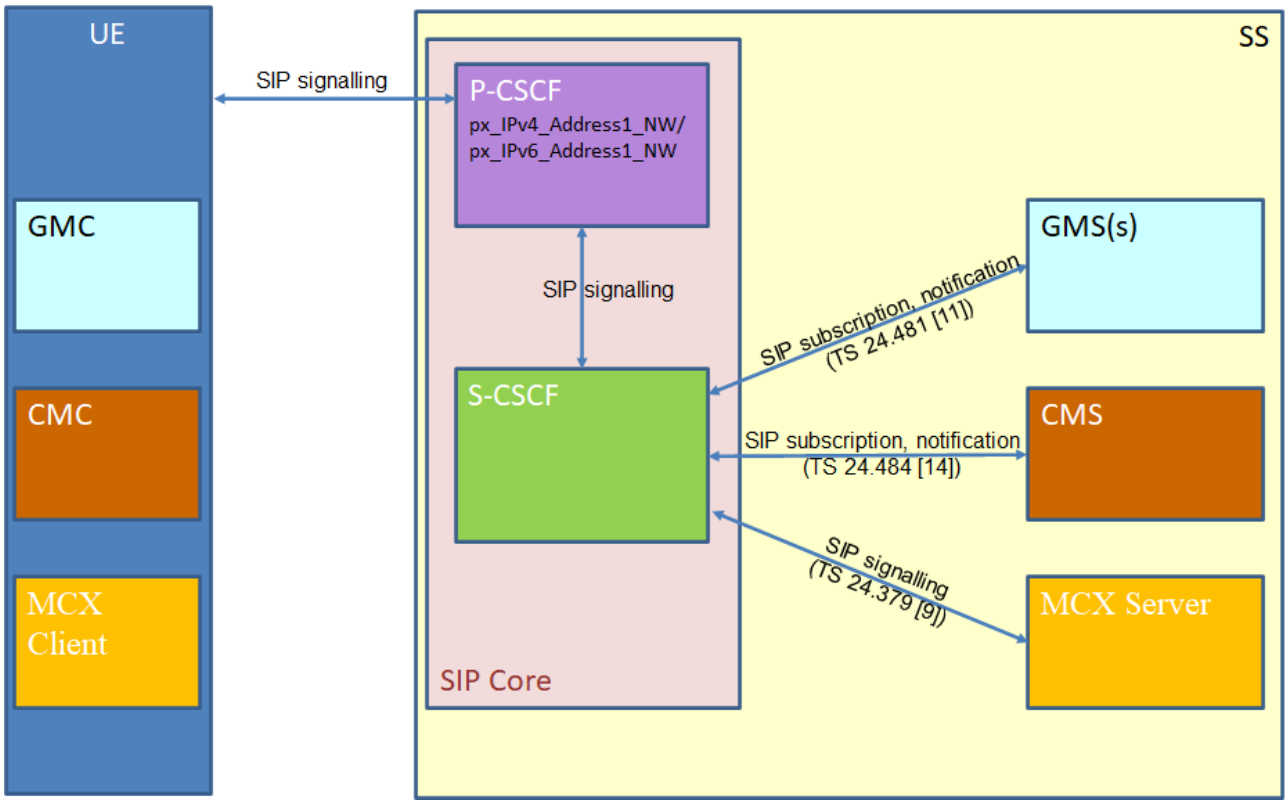



**Figure 7.2.2-1: Functional architecture for SIP-Signalling**

Regarding lower layer and IP signalling there are no differences compared to the IMS test model (TS 34.229-3 [28]).

The SIP signalling between the UE and the network entities is as specified in the test cases and according to common procedures and message content in TS 37.579-1 [2] with the following clarifications:

- Dialogs between the Group Management Client (GMC) and the Group Management Server (GMS)
  In general a GMC may subscribe to more than one GMS and TS 24.481 [11] shall be interpreted so that the GMC may even create more than one dialog with a single GMS.
  NOTE 1: In principle this applies for the Configuration Management Server too.
  NOTE 2: It is up to test case specification to deal with the possible scenarios and combinations of dialogs.

- Client Server Keys
  As not specified otherwise in the core specifications (TS 24.379 [9], TS 24.481 [11], TS 24.484 [14] and TS 33.180 [43]), there are no restriction for the UE regarding the CSK used in a dialog with the GMS(s), the CMS or the MCX server.
  ⇒ In general for CSK upload the UE may create and use different CSKs for the different servers and even for one and the same server the UE may use different CSKs for different dialogs. Furthermore a CSK download by the MCX server affects the communication between the MCX client and the MCX server only but not the CSKs used in any existing or future dialogs with a GMS or CMS; i.e. when the UE creates a dialog with a GMS or CMS after CSK download, it may use the downloaded CSK or any other CSK.

## 7.3 Interactive connectivity establishment (ICE) for pre-established session

According to TS 24.379 [9] for establishment of a pre-established session in general the MCPTT client shall gather ICE candidates according to RFC 5245 [63] and provide the candidates in the SDP offer of the initial SIP INVITE. Similar to the client the MCPTT server shall provide ICE candidates in its SDP answer and in general the client starts connectivity checks according to RFC 5245 [63].

Nevertheless, to reduce the test model requirements, the server side (at the system simulator) can be a lite implementation according to RFC 5245 [63] and as reflected in the default message contents for SDP Messages from the SS in TS 37.579-1 [2].

In addition - as working assumption - the UE shall be configured as lite implementation too.

⇒ When both ends are lite implementations none of them will start connectivity checks. Furthermore, there is no need for emulation for a STUN server for the client to gather ICE candidates as a lite implementation only includes host candidates for any media stream (RFC 5245 [63] clause 2.7).

**Conclusion:**
As long as there are no test requirements regarding ICE and as long as clients can be configured as lite implementations there are no ICE specific requirements for the test model.

# 8 Other SS requirements with TTCN-3 impact

## 8.1 Codec requirements

In addition to the requirements specified in TS 36.523-3 [27] clause 8.1, TS 38.523-3 [72] clause 8.1 and TS 34.229-3 [28] clause 7 the codec requirements of table 8.1-1 shall be applied.

**Table 8.1-1: Codec requirements**

| Type definitions | Codec requirements | Encoding rule in TTCN-3 |
|---|---|---|
| MIKEY_TypeDefs | RFC 3830 [64] clause 6 with extensions in RFC 6509 [65] and RFC 6043 [66] | "MIKEY-Codec" |
| MIKEY_MCX_Extension_TypeDefs | TS 33.180 [43] Annex E.6 | "Key Parameter Payload Types" |
| MCDataMessage_TypeDefs | TS 24.282 [57] clause 15 | "MCData Messages" |
| MCPTT_MediaControl_TypeDefs<br>FloorControl_Group<br>PreestablishedSessionControl_Group | <br>TS 24.380 [10] clause 8.2<br>TS 24.380 [10] clause 8.3 | <br>"FloorCtrlCodec"<br>"PreestablishedSessionCtrlCodec" |
| MCVideo_TransmissionControl_TypeDefs | TS 24.581 [56] clause 9 | "TransmissionCtrlCodec" |
| JSON | ES 201 873-11 [70] annex B, B.2 | "JSON" |

XML schemas are used according to ETSI ES 201 873-9 [75] with clarification provided in annex D.1.

NOTE: Basically, for XSD type definitions of different name spaces type compatibility according to clause 6.3 of ES 201 873-1 [31] is possible, i.e. XSD type definitions are compatible when their corresponding TTCN-3 type definitions are compatible. For compatible types common templates may be used in TTCN but it needs to be ensured that the appropriate encoding information is provided to the codec (e.g. by the context in which the common template is used).

## 8.2 External function definitions

In addition to the external functions specified in TS 36.523-3 [27] clause 8.2 and TS 38.523-3 [72] clause 8.2 there are the following MCX specific external functions:

| **TTCN-3 External Function** | | |
|---|---|---|
| **Name** | **fx_SAKKE_GeneratePublicKey** | |
| **Description** | Generate KMS public key (Z_T) for SAKKE (RFC 6508 [45] clause 2.2):<br>Z_T := [z_T]P | |
| **Parameters** | p_MasterSecret | master secret z_T (octetstring) |
| | p_ParameterSet | parameter set to be used; 1 per default, indicating use of parameter set 1 as defined in appendix A of RFC 6509 [46] |
| **Return Value** | octetstring | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_SAKKE_GenerateRSK** | |
| **Description** | Generate receiver secret key (RSK) for SAKKE (RFC 6508 [45] clause 2.2):<br>RSK := [(a + z_T)^-1]P<br>with 'a' being the identifier (UID) corresponding to the receiver's URI | |
| **Parameters** | p_MasterSecret | master secret z_T (octetstring) |
| | p_Identifier | UID for a given URI (octetstring) |
| | p_ParameterSet | parameter set to be used; 1 per default, indicating use of parameter set 1 as defined in appendix A of RFC 6509 [46] |
| **Return Value** | octetstring | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_SAKKE_EncapsulateKey** | |
| **Description** | Generate encapsulated data for SAKKE exchange according to RFC 6508 [45] | |
| **Parameters** | p_SSV | Shared secret value (octetstring): Key to be exchanged; according to 33.180 [43] E.1.1:<br>The GMK, PCK, CSK and MuSiK shall be 16 octets in length |
| | p_SakkePublicKey | SAKKE public key (octetstring) generated with fx_SAKKE_GeneratePublicKey |
| | p_UID | UID (octetstring) generated for the receiving entity's URI (in general the same URI as in IDRr payload of the MIKEY message carrying the encapsulated data) |
| | p_ParameterSet | parameter set to be used; 1 per default, indicating use of parameter set 1 as defined in appendix A of RFC 6509 [46] |
| **Return Value** | octetstring | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_SAKKE_ExtractKey** | |
| **Description** | Extract and validate 16 octet key from the encapsulated date for SAKKE exchange according to RFC 6508 [45] clause 6.2.2: if validation fails (TEST does not equal R_(b,S)) then omit shall be returned | |
| **Parameters** | p_EncapsulatedData | encapsulated data (octetstring) as received in the SAKKE payload of a MIKEY message |
| | p_SakkeRSK | receiver secret key (RSK) for SAKKE (octetstring) |
| | p_SakkePublicKey | SAKKE public key (octetstring) generated with fx_SAKKE_GeneratePublicKey |
| | p_UID | UID (octetstring) generated for the receiving entity's URI (should be the same URI as in IDRr payload of the MIKEY message carrying the encapsulated data |
| | p_ParameterSet | parameter set to be used; 1 per default, indicating use of parameter set 1 as defined in appendix A of RFC 6509 [46] |
| **Return Value** | template (omit) octetstring (16 octets if the key valid, omit otherwise) | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_ECCSI_GenerateKPAK** | |
| **Description** | Generate KMS Public Authentication Key (KPAK) for ECCSI (RFC 6507 [44] clause 4.2):<br>KPAK := [KSAK]G<br><br>The P-256 elliptic curve, base point and SHA-256 function as according to RFC 6509 [46] clause 2.1.1 shall be used. | |
| **Parameters** | p_KSAK | KMS Secret Authentication Key (KSAK): random secret non-zero integer modulo q (octetstring) |
| **Return Value** | octetstring | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_ECCSI_GenerateSskPvtPair** | |
| **Description** | Generate (SSK,PVT) Pair according to clause 5.1.1 of RFC 6507 [44]; if either the SSK or HS is zero modulo q (step 5 of the algorithm), the function shall return omit<br><br>The P-256 elliptic curve, base point and SHA-256 function as according to RFC 6509 [46] clause 2.1.1 shall be used. | |
| **Parameters** | p_UID | User ID (octetstring) |
| | p_KSAK | KMS Secret Authentication Key (KSAK) (octetstring) |
| | p_KPAK | KMS Public Authentication Key (KPAK) (octetstring) |
| | p_EphemeralValue | random (ephemeral) non-zero integer value v according to RFC 6507 [44] clause 5.1.1 |
| **Return Value** | template (omit) type record ECCSI_SskPvtPair_Type {<br>octetstring SSK,<br>octetstring PVT<br>} | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_ECCSI_SignMessage** | |
| **Description** | Sign a message according to RFC 6507 [44] clause 5.2.1:<br>return signature of the message or omit if HE + r * SSK is non-zero modulo q (step 4 of the algorithm)<br><br>The P-256 elliptic curve, base point and SHA-256 function as according to RFC 6509 [46] clause 2.1.1 shall be used. | |
| **Parameters** | p_Message | Message to be signed (octetstring) |
| | p_KPAK | KMS Public Authentication Key (KPAK) (octetstring) |
| | p_UID | Signer's User ID (octetstring) |
| | p_SSK | Secret Signing Key (octetstring) |
| | p_PVT | Public Validation Token (octetstring) |
| | p_EphemeralValue | random (ephemeral) non-zero integer value j according to RFC 6507 [44] clause 5.2.1 |
| **Return Value** | template (omit) octetstring | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_ECCSI_VerifySignature** | |
| **Description** | Verify a signature according to RFC 6507 [44] clause 5.2.2:<br>return true, when the signature is valid, false otherwise<br><br>The P-256 elliptic curve, base point and SHA-256 function as according to RFC 6509 [46] clause 2.1.1 shall be used. | |
| **Parameters** | p_Message | Message (octetstring) |
| | p_Signature | Message's signature (octetstring) |
| | p_KPAK | KMS Public Authentication Key (KPAK) (octetstring) |
| | p_UID | Signer's User ID (octetstring) |
| **Return Value** | boolean | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_XML_Encrypt** | |
| **Description** | Encrypt data<br>NOTE: the function is defined similar to openssl_encrypt and in principle it is independent from XML; nevertheless it is used for XML encrytion in context of MCX | |
| **Parameters** | p_Data | (octetstring) |
| | p_Method | type enumerated XML_EncryptionMethod_Type {<br>AES_128_GCM, // AES-GCM according to clause 5.2.4 of<br>// https://www.w3.org/TR/xmlenc-core1/ [41]<br>// with 96 bit Initialization Vector and<br>//128 bit Authentication Tag<br>AES_256_KEY_WRAP // according to RFC 3394 [47]<br>} |
| | p_Key | key for encryption (bitstring) |
| | p_IV | initial vector (octetstring) |
| **Return Value** | octetstring | In case of an authenticated encryption mechanism (e.g. AES_128_GCM) the Authentication Tag is added at the end of the encrypted data. |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_XML_Decrypt** | |
| **Description** | Decrypt data | |
| **Parameters** | p_Data | In case of an authenticated encryption mechanism (e.g. AES_128_GCM) the input data is the concatination of the encrypted data and Authentication Tag<br>(octetstring) |
| | p_Method | type enumerated XML_EncryptionMethod_Type {<br>AES_128_GCM, // AES-GCM according to clause 5.2.4 of<br>// https://www.w3.org/TR/xmlenc-core1/ [41]<br>// with 96 bit Initialization Vector and<br>//128 bit Authentication Tag<br>AES_256_KEY_WRAP // according to RFC 3394 [47]<br>} |
| | p_Key | key for encryption (bitstring) |
| | p_IV | initial vector (octetstring) |
| **Return Value** | octetstring | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_MCX_XML_AddEnvelopedSignature** | |
| **Description** | Add XML signature to the given XML document and return resulting XML document; according to W3C: "XML Signature Syntax and Processing (Second Edition)" [48]:<br><br>1. The given document has a Signature element with the name of the key to be used to sign the Signature's SignedInfo<br><br>2. Elements(s) to be signed: For elements which are addressed by a reference URI in the Signature's SignedInfo the DigestValue shall be generated and added to the corresponding Reference element of the Signature's SignedInfo<br><br>3. The SignedInfo shall be signed by generating the hash for the Signature's SignedInfo using the given key; this hash value shall be added to the Signature's SignatureValue. | |
| **Parameters** | p_XmlDocument | XML document (charstring) to be signed; the document contains all information to get signed with the given key:<br>- id(s) for the object(s) to be signed (e.g. KMS response)<br>- SignedInfo with reference(s) to objects to be signed within the document ("same-document" reference(s) according to RFC 3986 [53] clause 4.4), DigestAlgorithm, empty DigestValue |
| | p_Key | Key corresponding to the KeyName in the Signature's KeyInfo element (bitstring) |
| **Return Value** | universal charstring containing the document with all DigestValues and the signature of the SignedInfo | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_MCX_XML_CreateDetachedSignature** | |
| **Description** | Create XML signature for the given XML document and return resulting (encoded) XML signature (detached signature); according to TS 33.180 [43] clause 9.3.5, TS 24.379 [9] clause 6.6.3.1 and W3C: "XML Signature Syntax and Processing (Second Edition)" [48]: p_SignatureInfo contains the parameters needed to create the signature and to sign the document; the whole document gets signed irrespective of the Reference URI | |
| **Parameters** | p_XmlDocument | XML document to be signed |
| | p_XmldsigDocument | XML xmldsig document with signature element containing all information to sign the document (p_XmlDocument) |
| | p_Key | Key corresponding to the KeyName in the Signature's KeyInfo element |
| **Return Value** | universal charstring | XML xmldsig document with DigestValue and SignatureValue added to p_XmldsigDocument |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_XML_GetElementContent** | |
| **Description** | Return content of the node in an XML tree or XML branch as pointed at by the xpath expression | |
| **Parameters** | p_XmlBranch | XML tree (including the root element) or branch ot an XML tree |
| | p_XpathExpression | XCAP expression according to https://www.w3.org/TR/1999/REC-xpath-19991116/ |
| **Return Value** | universal charstring | string containing the content of the element or empty string if p_XpathExpression does not address an existing element of the given XML branch<br>NOTE: if the element may be empty, it may be necessary to check existence of the element with fx_XML_GetElement first |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_XML_SetElementContent** | |
| **Description** | Return XML tree or XML branch with replaced content of the node pointed at by the xpath expression | |
| **Parameters** | p_XmlBranch | XML tree (including the root element) or branch ot an XML tree |
| | p_XpathExpression | XCAP expression according to https://www.w3.org/TR/1999/REC-xpath-19991116/ |
| | p_NewContent | new content for the given node |
| **Return Value** | universal charstring | modified XML tree or XML branch |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_SHA_1** | |
| **Description** | Generic SHA-1 function | |
| **Parameters** | p_Data | data (octetstring) |
| **Return Value** | octetstring (hash value of 160 bits) | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_SHA_2** | |
| **Description** | Generic SHA-2 function | |
| **Parameters** | p_Function | type enumerated HASH_Function_Type {<br>SHA_256<br>// may be extended e.g SHA_224 etc.<br>} |
| | p_Data | data (octetstring) |
| **Return Value** | octetstring (representing 256 bits for SHA-256, 224 bits for SHA-224, ...) | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_JWK_Signature** | |
| **Description** | Generate JWK signature according to RFC 7515 [49] | |
| **Parameters** | p_String | string for which the signature shall be generated (charstring) |
| | p_Algorithm | algorithm to generate the hash:<br><br>type enumerated JWK_HashAlgorithm_Type { // RFC 7515 [49]<br>HS256, // HMAC SHA-256<br>RS256 // RSASSA-PKCS1-v1_5 SHA-256<br>} |
| | p_Key | key (octetstring) |
| **Return Value** | charstring (base64url encoded signature according to RFC 7515 [49]) | |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_XML_Log** | |
| **Description** | Log function for (unencrypted) XML documents | |
| **Parameters** | p_FileName | TTCN module from where the function is called |
| | p_LineNo | Line number from where the function is called |
| | p_UplinkDownlink | XML document has been received (UPLINK) or will be sent (DOWNLINK) |
| | p_XmlDocument | XML document |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_AEAD_Encrypt** | |
| **Description** | Encrypt data using AEAD algorithms according to RFC 5116 | |
| **Parameters** | p_Data | (octetstring) |
| | p_AssociatedData | (octetstring) |
| | p_Method | type enumerated AEAD_EncryptionMethod_Type {<br>AEAD_AES_128_GCM, // RFC 5116 clause 5.1<br>AEAD_AES_256_GCM // RFC 5116 clause 5.2<br>} |
| | p_Key | key for encryption (bitstring) |
| | p_IV | initial vector (octetstring) |
| **Return Value** | octetstring | Encrypted data appended by authentication tag |

| TTCN-3 External Function | | |
|---|---|---|
| **Name** | **fx_AEAD_Decrypt** | |
| **Description** | Decrypt data using AEAD algorithms according to RFC 5116 | |
| **Parameters** | p_Data | (octetstring) |
| | p_AssociatedData | (octetstring) |
| | p_Method | type enumerated AEAD_EncryptionMethod_Type {<br>AEAD_AES_128_GCM, // RFC 5116 clause 5.1<br>AEAD_AES_256_GCM // RFC 5116 clause 5.2<br>} |
| | p_Key | key for decryption (bitstring) |
| | p_IV | initial vector (octetstring) |
| **Return Value** | octetstring | Decrypted data |

# 9 IXIT Proforma

## 9.1 General

This partial IXIT proforma contained in the present document is provided for completion, when the related Abstract Test Suite is to be used against the Implementation Under Test (IUT).

Text in italics is a comment for guidance for the production of an IXIT, and is not to be included in the actual IXIT.

The completed partial IXIT will normally be used in conjunction with the completed ICS, as it adds precision to the information provided by the ICS.

# 9.2 MCX

## 9.2.1 MCX Client PIXIT

**Table 9.2.1-1: MCX Client Common PIXIT**

| Parameter Name | Parameter Type | Default Value | Supported Values | Description |
|---|---|---|---|---|
| **Client relevant IXIT** | | | | |
| px_MCX_Client_B_ID | charstring | "urn:uuid:cdcd34fb-c2f1-4488-a03a-6b5a5d360c47" | | MCX client ID for a remote user: As defined in TS 24.379 [9] clause 4.10 it is a UUID URN according to RFC 4122 [62], i.e. has the format "urn:uuid:XXXXXXXX-YYYY-ZZZZ-yyyy-zzzzzzzzzzzz" |
| **Users relevant IXIT** | | | | |
| px_MCPTT_ID_User_A | charstring | "mcptt-user-A-id@mcptt-op.gov" | NOTE 1 | MCPTT user identity (MCPTT ID) which is a globally unique identifier within the MCPTT service that represents the MCPTT user. Ref. TS 24.483 [13]. |
| px_MCVideo_ID_User_A | charstring | | NOTE 1 | MCVideo user identity (MCVideo ID) which is a globally unique identifier within the MCVideo service that represents the MCVideo user. Ref. TS 24.483 [13]. |
| px_MCData_ID_User_A | charstring | | NOTE 1 | MCData user identity (MCData ID) which is a globally unique identifier within the MCData service that represents the MCData user. Ref. TS 24.483 [13]. |
| px_MCPTT_User_A_Alias | charstring | "mcptt-user-A-alias" | | Alphanumeric alias of MCPTT user. Ref. TS 24.483 [13]. |
| px_MCVideo_User_A_Alias | charstring | "mcvideo-user-A-alias" | | Alphanumeric alias of MCVideo user. Ref. TS 24.483 [13]. |
| px_MCData_User_A_Alias | charstring | "mcdata-user-A-alias" | | Alphanumeric alias of MCData user. Ref. TS 24.483 [13]. |
| px_MCPTT_ID_FA_A | charstring | "mcptt-FA-A-id@mcptt-op.gov" | | The value is a "uri" attribute. TS 24.483 [13] clause 5.2.48W6 |
| px_MCData_ID_FA_A | charstring | "mcdata-FA-A-id@mcptt-op.gov" | | The value is a "uri" attribute. TS 24.483 [13] clause 10.2.97B3 |
| px_MCVideo_ID_FA_A | charstring | "mcvideo-FA-A-id@ mcptt-op.gov" | | The value is a "uri" attribute. TS 24.483 [13] clause 13.2.87A6 |
| px_MCX_User_A_ParticipantType | charstring | "first responder" | | Participant type of the MCPTT user. Ref. TS 24.483 [13]. |
| px_MCX_DomainName_Organization_A | charstring | "mcptt-op.gov" | | Domain name of the organization the user belongs to. |
| px_MCX_User_A_username | charstring | "MCPTT#U01" | | UE's User username used for user authentication |
| px_MCX_User_A_password | charstring | "psw@MCPTT&7" | | UE's User password used for user authentication |
| px_MCX_TokenRsaPrivateKey | octetstring | | | Private key being used to create signature for ID and Access Token |
| px_MCX_TrK | bitstring | | | Transport key for XML confidentiality and between the KMS and the MC KM client according to TS 33.180 [43] clause 9.3.3. |
| px_MCX_TrK_ID | B32_Type | | | Transport key ID for XML confidentiality and between the KMS and the MC KM client according to TS 33.180 [43] clause 9.3.3. |
| px_MCX_InK | bitstring | | | Integrity protection key for XML confidentiality and between the KMS and the MC KM client according to TS 33.180 [43] clause 9.3.3. |

| Parameter Name | Parameter Type | Default Value | Supported Values | Description |
|---|---|---|---|---|
| px_MCX_InK_ID | B32_Type | | | Integrity protection key ID for XML confidentiality and between the KMS and the MC KM client according to TS 33.180 [43] clause 9.3.3. |
| px_MCPTT_ID_User_B | charstring | "mcptt-user-B-id@ mcptt-op.gov" | | MCPTT user identity (MCPTT ID) which is a globally unique identifier within the MCPTT service that represents the MCPTT user. Ref. TS 24.483 [13]. |
| px_MCVideo_ID_User_B | charstring | "mcvideo-user-B-id@ mcptt-op.gov" | | MCVideo user identity (MCVideo ID) which is a globally unique identifier within the MCVideo service that represents the MCVideo user. Ref. TS 24.483 [13]. |
| px_MCData_ID_User_B | charstring | "mcdata-user-B-id@ mcptt-op.gov" | | MCData user identity (MCData ID) which is a globally unique identifier within the MCData service that represents the MCData user. Ref. TS 24.483 [13]. |
| px_MCX_User_B_ParticipantType | charstring | "first responder" | | Participant type of the MCX user. Ref. TS 24.483 [13]. |
| px_MCPTT_ID_User_C | charstring | "mcptt-user-C-id@ mcptt-op.gov" | | MCPTT user identity (MCPTT ID) which is a globally unique identifier within the MCPTT service that represents the MCPTT user. Ref. TS 24.483 [13]. |
| px_MCVideo_ID_User_C | charstring | "mcvideo-user-C-id@ mcptt-op.gov" | | MCVideo user identity (MCVideo ID) which is a globally unique identifier within the MCVideo service that represents the MCVideo user. Ref. TS 24.483 [13]. |
| px_MCData_ID_User_C | charstring | "mcdata-user-C-id@ mcptt-op.gov" | | MCData user identity (MCData ID) which is a globally unique identifier within the MCData service that represents the MCData user. Ref. TS 24.483 [13]. |
| px_MCX_User_C_ParticipantType | charstring | "first responder" | | Participant type of the MCX user. Ref. TS 24.483 [13]. |
| px_MCPTT_ID_User_D | charstring | "mcptt-user-D-id@ mcptt-op.gov" | | MCPTT user identity (MCPTT ID) which is a globally unique identifier within the MCPTT service that represents the MCPTT user. Ref. TS 24.483 [13]. |
| px_MCVideo_ID_User_D | charstring | "mcvideo-user-D-id@ mcptt-op.gov" | | MCVideo user identity (MCVideo ID) which is a globally unique identifier within the MCVideo service that represents the MCVideo user. Ref. TS 24.483 [13]. |
| px_MCData_ID_User_D | charstring | "mcdata-user-D-id@ mcptt-op.gov" | | MCData user identity (MCData ID) which is a globally unique identifier within the MCData service that represents the MCData user. Ref. TS 24.483 [13]. |
| px_MCX_SIP_PublicUserId_A_1 | charstring | | | (First) public user id of the SIP subscriber (user A) for MCX; the parameter shall be a SIP URI |

| Parameter Name | Parameter Type | Default Value | Supported Values | Description |
|---|---|---|---|---|
| px_MCX_SIP_PrivateUserId_A | charstring | | | Private user id of the SIP subscriber (user A) for MCX |
| px_MCX_SIP_PrivateUserId_B | charstring | | | Private user id of the SIP subscriber (user B) for MCX |
| px_MCX_SIP_HomeDomain_A | charstring | | | Home domain name of the SIP subscriber (user A) for MCX |
| px_MCX_SIP_PublicUserId_B | charstring | | | Public user id of user B |
| px_MCX_SIP_PublicUserId_C | charstring | | | Public user id of user C |
| px_MCX_SIP_RegistrationWithTemporaryIdentities | boolean | | | If true the UE shall derive the public user id, private user id and home domain name from the IMSI for SIP registration |
| px_MCData_IPConnectivityEndpointAddress_B | charstring | | | IP address of the IP connectivity endpoint at the emulated client B: either an IPv4 or an IPv6 address |
| **Groups relevant IXIT** | | | | |
| px_MCPTT_Group_A_ID | charstring | "mcptt-group-A@mcptt-op.gov" | | Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13]. |
| px_MCVideo_Group_A_ID | charstring | "mcvideo-group-A@mcptt-op.gov" | | Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13]. |
| px_MCData_Group_A_ID | charstring | "mcdata-group-A@mcptt-op.gov" | | Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13]. |
| px_MCPTT_Group_A_Name | charstring | "mcptt-group-A-name" | | A human readable Group name for the group |
| px_MCVideo_Group_A_Name | charstring | "mcvideo-group-A-name" | | A human readable Group name for the group |
| px_MCData_Group_A_Name | charstring | "mcdata-group-A-name" | | A human readable Group name for the group |
| px_MCX_Group_A_Owner_Organization | charstring | "mcx-op.gov" | | Indicates the group's owner organization the group belongs to. Ref. TS 24.483 [13]. |
| px_MCPTT_Group_A_preferred_VCodec | charstring | "AMR-WB" | | Preferred voice codec for the group (a RTP payload). MCPTT clients shall support the AMR-WB codec.<br>RFC 4566 [25]<br>TS 26.171 [26] |
| px_MCPTT_Group_B_ID | charstring | "mcptt-group-B@mcptt-op.gov" | | Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13]. |
| px_MCVideo_Group_B_ID | charstring | "mcvideo-group-B@mcptt-op.gov" | | Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13]. |
| px_MCData_Group_B_ID | charstring | "mcdata-group-B@mcptt-op.gov" | | Group ID for a group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13]. |

| Parameter Name | Parameter Type | Default Value | Supported Values | Description |
|---|---|---|---|---|
| px_MCPTT_Group_T_ID | charstring | "mcptt-group-T@mcptt-op.gov" | | Group ID for a temporary group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13]. |
| px_MCVideo_Group_T_ID | charstring | "mcvideo-group-T@mcptt-op.gov" | | Group ID for a temporary group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13]. |
| px_MCData_Group_T_ID | charstring | "mcdata-group-T@mcptt-op.gov" | | Group ID for a temporary group. Value is an "uri" attribute specified in OMA OMA-TS-XDM_Group-V1_1 that indicates the group id. Ref. TS 24.483 [13]. |
| **Miscellaneous IXIT** | | | | |
| px_MCX_APN | charstring | "mcx-apn" | | A single APN which the UE shall use to access each and all MCX relevant services.<br>The APN is provided in the initial UE configuration as specified in TS 37.579-1 [2] Table 5.5.8.1-1. |
| px_MCX_InitialRegistration_TypeOf1stPDN | MCX_Registration_PDN_Type | mcx | ims, internet, mcx | First PDN registered during initial registration (either 'ims' or 'internet' or 'mcx'; 'none' is not applicable as first PDN).<br>(NOTE 2) |
| px_MCX_InitialRegistration_TypeOf2ndPDN | MCX_Registration_PDN_Type | none | ims, internet, mcx, none | Second PDN registered during initial registration; in addition to 'ims' or 'internet' or 'mcx' it may be 'none' to indicate that there is no second PDN connectivity requested by the UE during initial registration.<br>(NOTE 2) |
| px_MCX_InitialRegistration_TypeOf3rdPDN | MCX_Registration_PDN_Type | none | ims, internet, mcx, none | Third PDN registered during initial registration; in addition to 'ims' or 'internet' or 'mcx' it may be 'none' to indicate that there is no third PDN connectivity requested by the UE during initial registration.<br>(NOTE 2) |
| px_MCX_CoordinateLatitude_Client_B | float | none | | Simulated latitude value sent to the UE to inform about Client_B location |
| px_MCX_CoordinateLongitude_Client_B | float | none | | Simulated longitude value sent to the UE to inform about Client_B location |
| px_MCX_CiphAlgo_Def | CiphAlgo | des_ede3_cbc | des_ede3_cbc, aes_cbc or nociph | Ciphering Algorithm for IPsec.<br><br>NOTE: Unless specified otherwise in the test prose „nociph" shall not be used for verification. |
| px_MCX_IPSecAlgorithm | IntAlgo | hmac_sha_1_96 | hmac_md5_96, hmac_sha_1_96 | Integrity Algorithm for IPsec |

NOTE 1: According to TS 23.280 [54] clause 8.1.2 a MC service ID shall be a URI; nevertheless in context of this specification only URIs consisting of userinfo, host and path elements (see RFC 3986 [53]) are supported.
Editor's note: A similar note may need to be added to other PIXITs too.

NOTE 2: Basically, there is no correlation to the TTCN PDN_Index_Type specifying PDN_1, PDN_2, etc. The mapping depends on the TTCN implementation for the respective IPCAN.

## 9.2.2 MCX Server PIXIT

**Table 9.2.2-1: MCX Server Common PIXIT**

| Parameter Name | Parameter Type | Default Value | Supported Values | Description |
|---|---|---|---|---|
| px_MCX_GroupCreationXUI | charstring | "mcx-gms@mcptt-op.gov" | | Indicates the group creation XUI information for creation of groups. Ref. TS 23.003 [21]. |
| px_MCX_IdMS_auth_IPAddress | charstring | | | IPv4/IPv6 address of the IdMS authorization endpoint |
| px_MCX_IdMS_auth_Port | integer | | | Port number of the IdMS authorization endpoint |
| px_MCX_IdMS_auth_Certificate | charstring | | | Identifier or file name of the certificate to be used during establishment of the TLS tunnel to the IdMS authorization endpoint |
| px_MCX_IdMS_token_IPAddress | charstring | | | IPv4/IPv6 address of the IdMS token endpoint |
| px_MCX_IdMS_token_Port | integer | | | Port number of the IdMS token endpoint |
| px_MCX_IdMS_token_Certificate | charstring | | | Identifier or file name of the certificate to be used during establishment of the TLS tunnel to the IdMS token endpoint |
| px_MCX_HTTP_Proxy_IPAddress | charstring | | | IPv4/IPv6 address of the HTTP proxy |
| px_MCX_HTTP_Proxy_Port | integer | | | Port number of the HTTP proxy |
| px_MCX_HTTP_Proxy_Certificate | charstring | | | Identifier or file name of the certificate to be used during establishment of the TLS tunnel to the HTTP proxy |
| px_MCX_InitialConfigServer_IPAddress | charstring | | | IPv4/IPv6 address of the server optionally being used to download the Initial UE Configuration document |
| px_MCX_InitialConfigServer_Port | integer | | | Port number of the server optionally being used to download the Initial UE Configuration document |
| px_MCX_InitialConfigServer_UriPath | charstring | "/cms/initial-ue-config" | | URI Path component: Absolute path used for HTTP requests addressing a server to download the UE initial configuration document |
| px_MCX_TLS_CipherSuite | TLS_CIPHER_Type | TLS_RSA_WITH_NULL_SHA | | Cipher suite to be used for TLS connections |
| px_MCX_OAuth_ClientId_A | charstring | | | Client ID of the UE's MCX application as used in OAuth signalling with the IdMS |
| px_MCX_OAuth_RedirectURI_A | charstring | | | Redirect URI used by the UE implementation (user agent, MCX client and OS) to redirect the OAuth authentication response to the MCX client application |

## 9.2.3 MCX Constant PIXIT Definitions

Several parameters for MCX conformance testing can be defined as constants as they are neither preconfigured at the UE nor at the SS. Table 9.2.3-1 lists these constants.

**Table 9.2.3-1: MCX Constant PIXIT Definitions**

| Constant Name | Constant Type | Value | Description |
|---|---|---|---|
| tsc_MCX_KMS_Hostname | charstring | "kms." & px_MCX_DomainName_Organization_A | FQDN of the KMS; used in initial UE configuration as domain name for the 'kms' URI in the App-Server-Info. |
| tsc_MCX_CMS_Hostname | charstring | "cms." & px_MCX_DomainName_Organization_A | FQDN of the CMS; used in initial UE configuration as domain name for the 'cms' URI in the App-Server-Info. |
| tsc_MCX_GMS_Hostname | charstring | "gms." & px_MCX_DomainName_Organization_A | FQDN of the GMS; used in initial UE configuration as domain name for the 'gms' URI in the App-Server-Info. |
| tsc_MCX_KMS_CertUri | charstring | "certificate1." & tsc_MCX_KMS_Hostname | Name of the KMS certificate sent to the UE during MCX user authentication |
| tsc_MCX_IdMS_auth_UriPath | charstring | "/idms/auth" | URI path to address the IdMS authorization endpoint |
| tsc_MCX_IdMS_token_UriPath | charstring | "/idms/token" | URI path to address the IdMS token endpoint |
| tsc_MCX_IdMS_userauth_UriPath | charstring | "/idms/userauth" | relative URI at IdMS used for user authentication |
| tsc_MCX_KMS_init_UriPath | charstring | "/keymanagement/identity/v1/init" | Path of the Request-URI for "KMS Initialize" request according to TS 33.180 [43] D.2.3 |
| tsc_MCX_KMS_keyprov_UriPath | charstring | "/keymanagement/identity/v1/keyprov" | Path of the Request-URI for "KMS KeyProvision" request according to TS 33.180 [43] D.2.4 |
| tsc_MCX_GMSURI | charstring | "subscription-proxy." & tsc_MCX_GMS_Hostname | used for <GMS-URI> element in the MCS UE initial configuration document according to TS 24.484 [14] clause 7.2.2.7 and TS 24.483 [13] clause 8.2.9; sub-domain of the GMS's hostname |
| tsc_MCX_CMSXCAPRootURI | charstring | "http://xcap." & tsc_MCX_CMS_Hostname | used for <CMS-XCAP-root-URI> element in the MCS UE initial configuration document according to TS 24.484 [14] clause 7.2.2.7 and TS 24.483 [13] clause 8.2.9C |
| tsc_MCX_GMSXCAPRootURI | charstring | "http://xcap." & tsc_MCX_GMS_Hostname | used for <GMS-XCAP-root-URI> element in the MCS UE initial configuration document according to TS 24.484 [14] clause 7.2.2.7 and TS 24.483 [13] clause 8.2.9B |
| tsc_MCPTT_PublicServiceId_A | charstring | "sip:" & "mcptt-server-A@" & px_MCX_DomainName_Organization_A | The URI of the MCPTT Server which is simulated by the SS |
| tsc_MCVideo_PublicServiceId_A | charstring | "sip:" & "mcvideo-server-A@" & px_MCX_DomainName_Organization_A | The URI of the MCVideo Server which is simulated by the SS |
| tsc_MCData_PublicServiceId_A | charstring | "sip:" & "mcdata-server-A@" & px_MCX_DomainName_Organization_A | The URI of the MCData Server which is simulated by the SS |

| Constant Name | Constant Type | Value | Description |
|---|---|---|---|
| tsc_MCPTT_PublicServiceId_B | charstring | "sip:" & "mcptt-server-B@" & px_MCX_DomainName_Organization_A | The URI of a second MCPTT Server which is implemented in the DUT used in MCPTT Server testing. |
| tsc_MCPTT_PublicServiceId_PF_A | charstring | "sip:" & "participating-mcptt-function-A@" & px_MCX_DomainName_Organization_A | The URI of the participating MCPTT function which configures the location reporting at the UE according to TS 24.379 [9] clause 13.2.2. |
| tsc_MCVideo_PublicServiceId_PF_A | charstring | "sip:" & "participating-mcvideo-function-A@" & px_MCX_DomainName_Organization_A | The URI of the participating MCVideo function which configures the location reporting at the UE according to TS 24.281 [55] clause 18.2.2. |
| tsc_MCData_PublicServiceId_PF_A | charstring | "sip:" & "participating-mcdata-function-A@" & px_MCX_DomainName_Organization_A | The URI of the participating MCData function which configures the location reporting at the UE according to TS 24.282 [57] clause 17.2.2. |
| tsc_MCX_SessionID_B | charstring | "sip:" & "sessionB@cf." & px_MCX_DomainName_Organization_A | The URI of the MCPTT session B identity. Ref. TS 24.483 [13]. SIP URI according to TS 24.379 [9] clause 4.5. |
| tsc_MCPTT_SessionId | charstring | "sip:" & "mcptt-session-A@cf." & px_MCX_DomainName_Organization_A | Service Id for MCPTT calls as used as Contact URI of the MCPTT server |
| tsc_MCVideo_SessionId | charstring | "sip:" & "mcvideo-session-A@cf." & px_MCX_DomainName_Organization_A | Service Id for MCVideo calls as used as Contact URI of the MCVideo server |
| tsc_MCData_SessionId | charstring | "sip:" & "mcdata-session-A@cf." & px_MCX_DomainName_Organization_A | Service Id for MCData calls as used as Contact URI of the MCData server |
| tsc_MCData_MSF_Hostname | charstring | "msf." & px_MCX_DomainName_Organization_A | Hostname of the media storage function (MSF) |
| tsc_MCData_MSF_URI | charstring | "http://" & tsc_MCData_MSF_Hostname & "/userA/files" | Absolute URI of the media storage function (MSF) |
| tsc_MCData_MsgSF_Hostname | charstring | "msgsf." & px_MCX_DomainName_Organization_A | Hostname of the message store function (MsgSF) |
| tsc_MCData_MsgSF_BaseURL | charstring | "/MsgStoreRoot/nms/v1/mcc160/UserA" | Base URL of the message store function (MsgSF) according to clause 5.2 of RESTful Network API for NMS [71] |
| tsc_MCData_MsgSF_AbsoluteBaseURL | charstring | "http://" & tsc_MCData_MsgSF_Hostname & tsc_MCData_MsgSF_BaseURL | Absolute Base URL of the message store function (MsgSF) |
| tsc_MCPTT_Group_A_ProSeLayer2GroupID | octetstring | '00000A'O | Indicates the Prose layer-2 group ID for the group. Ref. TS 23.303 [24]. |
| tsc_MCPTT_Group_D_ProSeLayer2GroupID | octetstring | '00000D'O | Indicates the Prose layer-2 group ID for the group. Ref. TS 23.303 [24]. |
| tsc_MCVideo_VSF_Hostname | charstring | "vsf." & px_MCX_DomainName_Organization_A | Hostname of the video storage function (VSF) where video files are stored |

| Constant Name | Constant Type | Value | Description |
|---|---|---|---|
| tsc_MCVideo_VSF_URI | charstring | "http://" & tsc_MCVideo_VSF_Hostname & "/userA/files" | Absolute URI of the message storage function (MSF) where video files are stored |
| tsc_MCData_NS_Hostname | charstring | "ns." & px_MCX_DomainName_Organization_A | Hostname of the notification server (NS) |
| tsc_MCData_NS_BaseURL | charstring | "/NSserverRoot/notificationchannel/v1" | Base URL of the notification server (NS) according to clause 5.2 of RESTful Network API for NC [76] |

## 9.3 MCX IPCAN test model

**Table 9.3-1: MCX IPCAN test model specific PIXIT**

| Parameter Name | Parameter Type | Default Value | Supported Values | Description |
|---|---|---|---|---|
| px_MCX_IPCAN_EpsBearerId_IMS | MCX_IPCAN_EpsBearerId_Type | 5 | 5..13 | Default EPS bearer for IMS PDN (if any) |
| px_MCX_IPCAN_EpsBearerId_INTERNET | MCX_IPCAN_EpsBearerId_Type | 6 | 5..13 | Default EPS bearer for Internet PDN (if any) |
| px_MCX_IPCAN_EpsBearerId_MCX | MCX_IPCAN_EpsBearerId_Type | 7 | 5..13 | Default EPS bearer for MCX PDN |
| px_MCX_IPCAN_DedicatedEpsBearerId_MCPTT | MCX_IPCAN_EpsBearerId_Type | 8 | 5..13 | Dedicated EPS bearer for MCPTT |
| px_MCX_IPCAN_DedicatedEpsBearerId_MCVideo | MCX_IPCAN_EpsBearerId_Type | 9 | 5..13 | Dedicated EPS bearer for MCVideo |
| px_MCX_IPCAN_DedicatedEpsBearerId_MCData | MCX_IPCAN_EpsBearerId_Type | 10 | 5..13 | Dedicated EPS bearer for MCData |

# 10 Postambles

## 10.1 Introduction

The purpose of the present clause 10 is to specify the postambles used to bring the UE to a well-defined state regardless of the UE state at the termination of main test body or of the SS conditions and values of the system information inherited from the test.

## 10.2 MCX

The postambles specified in TS 34.229-3 [28] are also applicable to MCX test cases.

# Annex A (normative): Test Suites

## A.1 Introduction

This annex references the approved Test Suites, which accompany the present document. The Test Suites have been produced using the Testing and Test Control Notation version 3 (TTCN-3) according to ES 201 873-1 [31].

## A.2 Baseline of specifications

Table A.2-1 lists the core specifications and test specifications, which the delivered Test Suites are based upon.

**Table A.2-1: References of the test and Core specifications**

| Type | Specification | Release | Version |
|---|---|---|---|
| **Core specifications** | TS 24.379 [9] | Note 1 | Note 2 |
| | TS 24.380 [10] | Note 1 | Note 2 |
| | TS 24.481 [11] | Note 1 | Note 2 |
| | TS 24.482 [12] | Note 1 | Note 2 |
| | TS 24.483 [13] | Note 1 | Note 2 |
| | TS 24.484 [14] | Note 1 | Note 2 |
| | TS 33.180 [43] | Note 1 | Note 2 |
| | TS 24.229 [16] | Note 1 | Note 2 |
| | TS 24.281 [55] | Note 1 | Note 2 |
| | TS 24.581 [56] | Note 1 | Note 2 |
| | TS 24.282 [57] | Note 1 | Note 2 |
| | TS 24.582 [58] | Note 1 | Note 2 |
| **Test specifications** | TS 37.579-1 [2] | Note 1 | Note 2 |
| | TS 37.579-2 [3] | Note 1 | Note 2 |
| | TS 37.579-4 [5] | Note 1 | Note 2 |
| | TS 37.579-6 [59] | Note 1 | Note 2 |
| | TS 37.579-7 [60] | Note 1 | Note 2 |
| NOTE 1: Latest release available, up to the release number of the present document.<br>NOTE 2: Latest available | | | |

## A.3 MCX Client Test Suites

### A.3.1 MCPTT Client Test Suites

Table A.3.1-1 lists all approved test cases from TS 37.579-2 [3].

For a given test case, the following variants are distinguished (if applicable):

- IPCAN 4G: MCX IPCAN 4G test model.
- EUTRA: MCX EUTRA test model.
- IPCAN 5G: MCX IPCAN 5G test model.
- NR5GC: MCX NR5GC test model.

An "X" in columns IPCAN 4G, EUTRA, IPCAN 5G or NR5GC indicates the test case is approved for the respective variant.

An "-" in columns IPCAN 4G, EUTRA, IPCAN 5G or NR5GC indicates the test case is not applicable to the respective variant.

**Table A.3.1-1: MCPTT Client TTCN test cases from TS 37.579-2 [3]**

| Test case | Description | IPCAN 4G | EUTRA | IPCAN 5G | NR5GC |
|---|---|---|---|---|---|
| 1.MCPTT | MCPTT / Configuration / Authentication / User Authorisation / UE Configuration / User Profile | X | X | | X |
| 3.MCPTT | MCPTT / Configuration / Group Affiliation / Remote change / De-affiliation / Home MCPTT system | X | X | | X |
| 4.MCPTT | MCPTT / Configuration / Pre-established Session Establishment / Pre-established Session Modification / Pre-established Session Release | X | | | |
| 6.MCPTT | Configuration / Download CSK | X | | | |
| 8.MCPTT | Configuration / Functional Alias / Functional alias status determination / Activate functional alias / Deactivate functional alias | X | X | | X |
| 1.1.1.MCPTT | MCPTT / On-network / On-demand Pre-arranged Group Call / Automatic Commencement Mode / Floor Control / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group Call / Cancel Imminent Peril State / Client Originated (CO) | X | | | |
| 1.1.2.MCPTT | MCPTT / On-network / On-demand Pre-arranged Group Call / Automatic Commencement Mode / Floor Control / Upgrade to Emergency Group Call / Cancel Emergency State / Upgrade to Imminent Peril Group Call / Cancel Imminent Peril State / Client Terminated (CT) | X | | | |
| 1.1.3.MCPTT | MCPTT / On-network / On-demand Pre-arranged Group Call / Manual Commencement Mode / Client Originated (CO) | X | | | |
| 1.1.4.MCPTT | MCPTT / On-network / On-demand Pre-arranged Group Call / Manual Commencement Mode / Client Terminated (CT) | X | | | |
| 1.1.5.MCPTT | MCPTT / On-network / Pre-arranged Group Call using pre-established session / Client originated Pre-established Session Release with associated MCPTT session / Client Originated (CO) | X | | | |
| 1.1.6.MCPTT | MCPTT / On-network / Pre-arranged Group Call using pre-established session / Automatic Commencement Mode / Server originated Pre-established Session Release with associated MCPTT session / Client Terminated (CT) | X | | | |
| 1.1.8.MCPTT | MCPTT / On-network / Pre-arranged Broadcast Group Call / Client Originated (CO) | X | | | |
| 1.1.9.MCPTT | MCPTT / On-network / Pre-arranged Broadcast Group Call / Client Terminated (CT) | X | | | |
| 1.1.10.MCPTT | MCPTT / On-network / Broadcast Group Call with Temporary Group / Client Originated (CO) | X | | | |
| 1.1.11.MCPTT | MCPTT / On-network / Pre-arranged Emergency Group Call / Client Originated (CO) | X | | | |
| 1.1.12.MCPTT | MCPTT / On-network / Pre-arranged Emergency Group Call / Client Terminated (CT) | X | | | |
| 1.1.13.MCPTT | MCPTT / On-network / Pre-Arranged Imminent Peril Group Call / Client Originated (CO) | X | | | |
| 1.1.14.MCPTT | MCPTT / On-network / Pre-Arranged Imminent Peril Group Call / Client Terminated (CT) | X | | | |
| 1.1.16.MCPTT | MCPTT / On-network / Emergency Alert / Client Terminated (CT) | X | X | | X |
| 1.1.17.MCPTT | On-network / Broadcast Group Call using pre-established session / Client originated Pre-established Session Release with associated MCPTT session / Client Originated (CO) | X | | | |
| 1.1.18.MCPTT | MCPTT / On-network / Broadcast Group Call using pre-established session / Client originated Pre-established Session Release with associated MCPTT session / Client Terminated (CT) | X | | | |
| 1.1.19.MCPTT | On-network / On-demand Pre-arranged Group Call / Active functional alias / Client Originated (CO) | X | | | |
| 1.1.20.MCPTT | On-network / On-demand Pre-arranged Group Call / Multi Talker | X | | | |
| 1.1.21.MCPTT | On-network / On-demand Pre-arranged Group Call / No Implicit Floor Control / Client Originated (CO) | X | | | |
| 1.2.7.MCPTT | MCPTT / On-network / Chat Group Call / Emergency Group Call / Client Originated (CO) | X | | | |
| 1.2.8.MCPTT | MCPTT / On-network / Chat Group Call / Emergency Group Call / Client Terminated (CT) | X | | | |
| 1.2.9.MCPTT | MCPTT / On-network / Chat Group Call / Imminent Peril Group Call / Client Originated (CO) | X | | | |
| 1.2.10.MCPTT | MCPTT / On-network / Chat Group Call / Imminent Peril Group Call / Client Terminated (CT) | X | | | |
| 1.2.11.MCPTT | MCPTT / On-network / Chat Group Call / Join Chat Group Session / Upgrade to Emergency / Cancel Emergency / Upgrade to Imminent Peril / Cancel Imminent Peril / Client Originated (CO) | X | | | |
| 1.2.12.MCPTT | MCPTT / On-network / Chat Group Call / Upgrade to Emergency / Cancel Emergency / Upgrade to Imminent Peril / Cancel Imminent Peril / Client Originated (CT) | X | | | |
| 1.2.13.MCPTT | On-network / Chat Group Call / Join Chat Group Session / Active functional alias / Client Originated (CO) | X | | | |

| | | | | | |
|---|---|---|---|---|---|
| 2.1.MCPTT | MCPTT / On-network / Private Call / On-demand / Automatic Commencement Mode / With Floor Control confidentiality and integrity protection/ Upgrade to Emergency Call / Cancellation of Emergency on User request / Client Originated (CO) | X | | | |
| 2.2.MCPTT | MCPTT / On-network / Private Call / On-demand / Automatic Commencement Mode / With Floor Control confidentiality and integrity protection/ Upgrade to Emergency Call / Cancellation of Emergency on User request / Client Terminated (CT) | X | | | |
| 2.3.MCPTT | MCPTT / On-network / Private Call / On-demand / Automatic Commencement Mode / Without Floor Control / Client Originated (CO) | X | | | |
| 2.4.MCPTT | MCPTT / On-network / Private Call / On-demand / Automatic Commencement Mode / Without Floor Control / Client Terminated (CT) | X | | | |
| 2.5.MCPTT | MCPTT / On-network / Private Call / Emergency Private Call / On-demand / Automatic Commencement Mode / Force of automatic commencement mode / Without Floor Control / Client Originated (CO) | X | | | |
| 2.6.MCPTT | MCPTT / On-network / Private Call / Emergency Private Call / On-demand / Manual Commencement Mode / Force of automatic commencement mode / Without Floor Control / Client Terminated (CT) | X | | | |
| 2.7.MCPTT | MCPTT / On-network / Private Call / On-demand / Manual Commencement Mode / Without Floor Control / Client Originated (CO) | X | | | |
| 2.8.MCPTT | MCPTT / On-network / Private Call / On-demand / Manual Commencement Mode / Without Floor Control / Client Terminated (CT) | X | | | |
| 2.14.MCPTT | On-network / Private Call / Ambient listening call / Remotely initiated Ambient listening call / Remotely initiated ambient listening call release / Success / Client Originated (CO) / Server initiated ambient call release | X | | | |
| 2.15.MCPTT | MCPTT / On-network / Private Call / Ambient listening call / Remotely initiated Ambient listening call / Remotely initiated ambient listening call release / Success / Client Terminated (CT) | X | | | |
| 2.16.MCPTT | MCPTT / On-network / Private Call / Ambient listening call / Locally initiated Ambient listening call / Locally initiated ambient listening call release / Success / Client Originated (CO) / Server initiated ambient call release | X | | | |
| 2.17.MCPTT | MCPTT / On-network / Private Call / Ambient listening call / Locally initiated Ambient listening call / Locally initiated ambient listening call release / Success / Client Terminated (CT) | X | | | |

## A.3.2 MCVideo Client Test Suites

None.

## A.3.3 MCData Client Test Suites

Table A.3.3-1 lists all approved test cases from TS 37.579-7 [60].

For a given test case, the following variants are distinguished (if applicable):

- IPCAN 4G: MCX IPCAN 4G test model.
- EUTRA: MCX EUTRA test model.
- IPCAN 5G: MCX IPCAN 5G test model.
- NR5GC: MCX NR5GC test model.

An "X" in columns IPCAN 4G, EUTRA, IPCAN 5G or NR5GC indicates the test case is approved for the respective variant.

An "-" in columns IPCAN 4G, EUTRA, IPCAN 5G or NR5GC indicates the test case is not applicable to the respective variant.

**Table A.3.3-1: MCData Client TTCN test cases from TS 37.579-7 [60]**

| Test case | Description | IPCAN 4G | EUTRA | IPCAN 5G | NR5GC |
|---|---|---|---|---|---|
| 5.1.MCData | Configuration / Authentication / User Authorization / UE Configuration / User Profile / Key Generation | X | | | |
| 6.1.1.MCData | On-network / Short Data Service (SDS) / Standalone SDS Using Signaling Control Plane / One-to-one Standalone SDS / Client Originated (CO) | X | | | |
| 6.1.3.MCData | On-network / Short Data Service (SDS) / Standalone SDS Using Signaling Control Plane / Group Standalone SDS / Client Originated (CO) | X | | | |
| 6.1.4.MCData | On-network / Short Data Service (SDS) / Standalone SDS Using Signaling Control Plane / Group Standalone SDS / Client Terminated (CT) | X | | | |

# Annex B (informative): Style Guide

## B.1 Introduction

The style guide specified in TS 36.523-3 [27] Annex B applies to the present document.

# Annex C (normative): Common TTCN-3 Definitions

## C.0 Introduction

The present Annex C specifies the TTCN-3 type definitions used at the system interface to configure and control the SS. The type definitions are common, independent from test model.

In case of discrepancy between the content of the present Annex C and the equivalent TTCN-3 definitions / semantic requirements found in the TTCN modules provided as attachments to the present specification, the latter shall take precedence.

NOTE: This annex is automatically generated from the TTCN-3 modules provided as attachment to the present specification and containing the listed TTCN-3 type definitions.

## C.1 SRTP_ASP_TypeDefs

### C.1.1 Type_Definitions

**SRTP_EncryptionAlgorithm_Type**

| TTCN-3 Enumerated Type | |
|---|---|
| **Name** | **SRTP_EncryptionAlgorithm_Type** |
| **Comment** | |
| Null | RFC 3830 clause 6.10.1 |
| AES_CM | RFC 3830 clause 6.10.1 |
| AES_F8 | RFC 3830 clause 6.10.1 |
| AES_GCM | RFC 7714 clause 14.3 |

**SRTP_AuthenticationAlgorithm_Type**

| TTCN-3 Enumerated Type | |
|---|---|
| **Name** | **SRTP_AuthenticationAlgorithm_Type** |
| **Comment** | |
| Null | RFC 3830 clause 6.10.1 |
| HMAC_SHA_1 | RFC 3830 clause 6.10.1 |
| RCCm1 | RFC 4771 clause 4 |
| RCCm2 | RFC 4771 clause 4 |
| RCCm3 | RFC 4771 clause 4 |

**SRTP_PRF_Type**

| TTCN-3 Enumerated Type | |
|---|---|
| **Name** | **SRTP_PRF_Type** |
| **Comment** | |
| AES_CM | RFC 3830 clause 6.10.1 |

### SRTP_SecurityProfile_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **SRTP_SecurityProfile_Type** | | |
| **Comment** | parameters for SRTP security profiles according to 33.180 annex E:<br>- Table E.2.2-1 (RTP group call: GMK)<br>- Table E.3.2-1 (RTP private call: PCK)<br>- Table E.4.2-1 (RTCP: CSK and MuSiK)):<br>The parameters may be contained/negotiated in MIKEY Security Policy Payload (RFC 3830) | | |
| EncryptionAlgorithm | SRTP_EncryptionAlgorithm_Type | | SRTP type 0; RFC 3830 clause 6.10.1 |
| SessionEncryptionKeyLength | integer | | SRTP type 1; RFC 3830 clause 6.10.1 |
| AuthenticationAlgorithm | SRTP_AuthenticationAlgorithm_Type | opt | SRTP type 2; RFC 3830 clause 6.10.1 |
| SessionSaltKeyLength | integer | | SRTP type 4; RFC 3830 clause 6.10.1 |
| PRF | SRTP_PRF_Type | | SRTP type 5; RFC 3830 clause 6.10.1 |
| KeyDerivationRate | integer | | SRTP type 6; RFC 3830 clause 6.10.1 |
| ROC_TransmissionRate | integer | opt | SRTP type 13; RFC 4771 clause 4 |
| SRTP_AuthenticationTagLength | integer | opt | SRTP type 18; RFC 4771 clause 4 |
| SRTCP_AuthenticationTagLength | integer | opt | SRTP type 19; RFC 4771 clause 4 |
| AEAD_AuthenticationTagLength | integer | | SRTP type 20; RFC 7714 clause 14.3 |

### SRTP_KeyInfo_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **SRTP_KeyInfo_Type** | | |
| **Comment** | SRTP Master Key, SRTP Master Salt and MKI according to 33.180 clauses 7.4.1, 7.4.2 and 9.4.6, RFC 3830 and RFC 3711 | | |
| MasterKey | octetstring | | SRTP master key according to RFC 3711 (128 bits) |
| MasterSalt | octetstring | | SRTP master salt according to RFC 3711 and RFC 7714 (96 bits):<br>33.180 clauses 7.5 and 9.4.8 specify the AEAD_AES_128_GCM algorithm as defined in RFC 7714 to be used for confidentiality and data authentication of the media stream and the media control data.<br>RFC 7714 specifies in clause 12 a master salt length of 96 and in clause 11 that the AES_CM PRF KDF according to RFC 3711 shall be used.<br>Nevertheless RFC 3711 clause 8.2 requires a master salt length of 112.<br>This contradiction is resolved by the RFC Errata for RFC 7714:<br>The master salt shall be 96 bit in the first place (as according to RFC 7714) but then multiplied by 2^16 to fit to the requirements of RFC 3711 (112 bit).<br>=> MasterSalt provides a 96-bit value which needs to be padded by the system adaptor to get the 112-bit master salt as needed according to RFC 3711 |
| MKI | octetstring | opt | SRTP Master Key Identifier: if not present => no MKI to be added to the SRTP/SRTCP packet |

### SRTP_Config_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **SRTP_Config_Type** | | |
| **Comment** | SRTP configuration parameters | | |
| KeyInfo | SRTP_KeyInfo_Type | | |
| SecurityProfile | SRTP_SecurityProfile_Type | | |

### SRTP_Connection_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **SRTP_Connection_Type** | | |
| **Comment** | configuration parameters for a single (UDP) connection | | |
| Connection | IP_Connection_Type | | RTP/RTCP configuration shall be applied for packets matching the given Connection:<br>in general a connection is given by<br>Protocol := udp<br>Local := { IP address of the SS, port number at the SS }<br>Remote := omit<br>(but other matching criteria may be used if needed)<br>NOTE: TTCN shall ensure that different connection can be identified without ambiguity (in general even the local port number should be sufficient) |
| LocalSSRC | O4_Type | | SSRC of the SS |
| SRTP_Config | SRTP_Config_Type | opt | configuration of SRTP/SRTCP for the given connection; if not present => no SRTP/SRTCP |

### SRTP_ConnectionTable_Type

| TTCN-3 Record of Type | |
|---|---|
| **Name** | **SRTP_ConnectionTable_Type** |
| **Comment** | |
| record of SRTP_Connection_Type | |

### SRTP_CTRL_REQ

| TTCN-3 Union Type | | |
|---|---|---|
| **Name** | **SRTP_CTRL_REQ** | |
| **Comment** | configuration ASP at the system interface | |
| Config | SRTP_ConnectionTable_Type | table of all connection to be configured for SRTP/SRTCP (or RTP/RTCP);<br>the configuration overrides any previous configuration; it is not foreseen to add, modify or remove single connections |

### SRTP_CTRL_IND

| TTCN-3 Union Type | | |
|---|---|---|
| **Name** | **SRTP_CTRL_IND** | |
| **Comment** | ASP at the system interface for system confirmations and indications | |
| Confirm | Null_Type | confirmation for configuration |
| Error | Null_Type | indication of errors |

### C.1.2 System_Interface

**SRTP_PORT**

| TTCN-3 Port Type | | |
|---|---|---|
| **Name** | **SRTP_PORT** | |
| **Comment** | | |
| out | SRTP_CTRL_REQ | |
| in | SRTP_CTRL_IND | |

## C.2 IP_ASP_TypeDefs

General Notes:
NOTE 1:
In general the handling of IP data shall be independent from the RAT being used on lower layers.
NOTE 2:
It shall be possible for SS implementation to reuse existing IP stack implementations in the system adaptor;
therefore the well-known concept of socket programming shall be supported
(regardless of whether those are used in the system adaptor implementation or not)
NOTE 3:
Since in general at the network side there are several different IP addresses the SS needs to simulate more than one IP address;
that can be based on a concept of multiple virtual network adaptors
NOTE 4:
There is no easy way to control the routing of IP data for an IP connection from above the IP stack
i.e. there are no parameters at the socket interface to determine e.g. cell id and DRB id
=> another independent logical entity (DRB-MUX) is needed below the IP stack which is responsible to control the routing of IP packets from/to DRBs in different cells of different RATs

Reference:
An introduction to socket programming can be found in
UNIX Network Programming Volume 1, Third Edition: The Sockets Networking API
by W. Richard Stevens, Bill Fenner, Andrew M. Rudoff

### C.2.1 IP_Common

**IP_Common: Basic Type Definitions**

| TTCN-3 Basic Types | | |
|---|---|---|
| **PortNumber_Type** | UInt16_Type | |

**IPv4_AddrInfo_Type**

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **IPv4_AddrInfo_Type** | | |
| **Comment** | IPv4 specific info of the socket addr (AF_INET) | | |
| Addr | charstring | | IP Address as string (IP v4 dot notation) to be converted to 32-bit unsigned integer |

### IPv6_AddrInfo_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **IPv6_AddrInfo_Type** | | |
| **Comment** | IPv6 specific info of the socket addr (AF_INET6);<br>NOTE: sin6_flowinfo can be ignored and set to 0 | | |
| Addr | charstring | | to be converted to sin6_addr |
| ScopeId | UInt32_Type | opt | sin6_scope_id<br>in general an IPv6 address is like "fe80::1%eth0" with eth0 being the network adaptor mapped to a scope id (Unix)<br>assumption:<br>for UE conformance testing it is not necessary to distinguish different scopes and the scope id in general can be determined by the system adaptor => omit |

### IP_AddrInfo_Type

| TTCN-3 Union Type | | |
|---|---|---|
| **Name** | **IP_AddrInfo_Type** | |
| **Comment** | | |
| V4 | IPv4_AddrInfo_Type | |
| V6 | IPv6_AddrInfo_Type | |

### IP_Socket_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **IP_Socket_Type** | | |
| **Comment** | Socket | | |
| IpAddr | IP_AddrInfo_Type | opt | IP address |
| Port | PortNumber_Type | opt | port number |

### InternetProtocol_Type

| TTCN-3 Enumerated Type | |
|---|---|
| **Name** | **InternetProtocol_Type** |
| **Comment** | |
| udp | |
| tcp | |
| icmp | |
| icmpv6 | |

### IP_Connection_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **IP_Connection_Type** | | |
| **Comment** | A connection between peer-to-peer entities is unambiguously defined by the protocol (udp/tcp/icmp/icmpv4), the local socket and the remote socket | | |
| Protocol | InternetProtocol_Type | | |
| Local | IP_Socket_Type | opt | |
| Remote | IP_Socket_Type | opt | |

# C.3 CommonDefs

**CommonDefs: Constant Definitions**

| TTCN-3 Basic Types | | | |
|---|---|---|---|
| **tsc_UInt16Max** | integer | 65535 | |
| **tsc_UInt32Max** | integer | 4294967295 | |

**CommonDefs: Basic Type Definitions**

| TTCN-3 Basic Types | | |
|---|---|---|
| **O4_Type** | octetstring length(4) | |
| **Null_Type** | boolean (true) | dummy type for 'typeless' fields in unions |
| **UInt16_Type** | integer (0 .. tsc_UInt16Max) | |
| **UInt32_Type** | integer (0 .. tsc_UInt32Max) | |

# C.4 References to TTCN-3

| References to TTCN-3 | | |
|---|---|---|
| **SRTP_ASP_TypeDefs** | Common/SRTP_ASP_TypeDefs.ttcn | Rev 29933 |
| **IP_ASP_TypeDefs** | IP_PTC/IP_ASP_TypeDefs.ttcn | Rev 27511 |
| **CommonDefs** | Common/CommonDefs.ttcn | Rev 29871 |

# Annex D (normative): SIP Type Definitions and XSD References

## D.1 XML Schema Definitions (XSD)

Common XML schema definitions according to TS 34.229-3 [28] Table G.0.1-1 are used. In addition, there are the MCX specific XML schema definitions as according to table D.1-1.

**Table D.1-1: MCX specific definitions**

| XML Schema (XSD) | Source | Name space |
|---|---|---|
| RFC4354-poc-settings | RFC 4354 [52] clause 6.1 | urn:oma:params:xml:ns:poc:poc-settings |
| RFC4661-SimpleFilter | RFC 4661 [34] clause 7 | urn:ietf:params:xml:ns:simple-filter |
| RFC5261-patch-ops (NOTE 2) | RFC 5261 [50] clause 8 | (NOTE 1) |
| RFC5874-xcap-diff | RFC 5874 [51] clause 4 | urn:ietf:params:xml:ns:xcap-diff |
| IANA-resource-lists | RFC 4826 [34] clause 3.2 https://www.iana.org/assignments/xml-registry/schema/resource-lists.xsd | urn:ietf:params:xml:ns:resource-lists |
| poc_listService-v1_0 | https://www.openmobilealliance.org/tech/profiles/poc_listService-v1_0.xsd | urn:oma:xml:poc:list-service |
| xdm_extensions-v1_0 | https://www.openmobilealliance.org/tech/profiles/xdm_extensions-v1_0.xsd | urn:oma:xml:xdm:extensions |
| xdm_rsrclst_uriusage-v1_0 | https://www.openmobilealliance.org/tech/profiles/xdm_rsrclst_uriusage-V1_0.xsd | urn:oma:xml:xdm:resource-list:oma-uriusage |
| rest_netapi_common-v1_0 | https://www.openmobilealliance.org/tech/profiles/rest_netapi_common-v1_0.xsd | urn:oma:xml:rest:netapi:common:1 |
| rest_netapi_nms-v1_0 (NOTE 7) | https://www.openmobilealliance.org/tech/profiles/rest_netapi_nms-v1_0.xsd | urn:oma:xml:rest:netapi:nms:1 |
| rest_netapi_notificationchannel-v1_0 | https://www.openmobilealliance.org/tech/profiles/rest_netapi_notificationchannel-v1_0.xsd | urn:oma:xml:rest:netapi:notificationchannel:1 |
| xenc-schema | W3C [41] | http://www.w3.org/2001/04/xmlenc#' |
| xmldsig-core-schema | W3C [42] | http://www.w3.org/2000/09/xmldsig# |
| TS24281_mcvideoinfo (NOTE 3) | TS 24.281 [55] Annex F.1.2 | urn:3gpp:ns:mcvideoInfo:1.0 |
| TS24281_mcvideolocationinfo | TS 24.281 [55] Annex F.3.2 | urn:3gpp:ns:mcvideoLocationInfo:1.0 |
| TS24281_mcvideoPresInfo | TS 24.281 [55] Table 8.3.1.2-1 | urn:3gpp:ns:mcvideoPresInfo:1.0 |
| TS24281_mcvideoPresInfoFA | TS 24.281 [55] Table 20.3.1.2-1 | urn:3gpp:ns:mcvideoPresInfoFA:1.0 |
| TS24281_preconfiguredRegroup | TS 24.281 [55] Annex F.6.2 | urn:3gpp:ns:preconfiguredRegroup:1.0 NOTE 6 |
| TS24282_mcdatainfo | TS 24.282 [57] Annex D.1.2 | "urn:3gpp:ns:mcdataInfo:1.0" |
| TS24282_mcdatalocationinfo | TS 24.282 [57] Annex D.4.2 | urn:3gpp:ns:mcdataLocationInfo:1.0 |
| TS24282_mcdataPresInfo | TS 24.282 [57] Table 8.4.1.2-1 | "urn:3gpp:ns:mcdataPresInfo:1.0" |
| TS24282_mcdataPresInfoFA | TS 24.282 [57] Table 22.3.1.2-1 | "urn:3gpp:ns:mcdataPresInfoFA:1.0" |
| TS24282_preconfiguredRegroup | TS 24.282 [57] Annex F.6.2 | urn:3gpp:ns:preconfiguredRegroup:1.0 NOTE 6 |
| TS24379_mcpttaff (NOTE 4) | TS 24.379 [9] Annex F.4.2 TS 24.281 [55] Annex F.4.2 TS 24.282 [57] Annex D.3.2 | urn:3gpp:ns:affiliationCommand:1.0 |
| TS24379_mcpttinfo | TS 24.379 [9] Annex F.1.2 | urn:3gpp:ns:mcpttInfo:1.0 |
| TS24379_mcpttlocationinfo | TS 24.379 [9] Annex F.3.2 | urn:3gpp:ns:mcpttLocationInfo:1.0 |
| TS24379_mcpttsigneddoc | TS 24.379 [9] Annex F.6.2 | urn:3gpp:ns:mcpttSignedDoc:1.0 |
| TS24379_preconfiguredRegroup | TS 24.379 [9] Annex F.7.2 | urn:3gpp:ns:preconfiguredRegroup:1.0 NOTE 6 |
| TS24379_mcpttPresInfo | TS 24.379 [9] Table 9.3.1.2-1 | urn:3gpp:ns:mcpttPresInfo:1.0 |
| TS24379_mcpttPresInfoFA | TS 24.379 [9] Table 9A.3.1.2-1 | urn:3gpp:ns:mcpttPresInfoFA:1.0 |
| TS24379_poc-settings (NOTE 5) | TS 24.379 [9] Table 7.4.1.2.2-2 TS 24.281 [55] Table 7.4.1.2.2-2 | urn:3gpp:mcsSettings:1.0 |
| TS24481-mcptt-group | TS 24.481 [11] clause 7.2.4.2 | urn:3gpp:ns:mcpttGroupInfo:1.0 |
| TS24481-mcpttGMOP | TS 24.481 [11] clause 7.3.3 | urn:3gpp:ns:mcpttGMOP:1.0 |
| TS24481-mcpttGKTP | TS 24.481 [11] clause 7.7.4.2 | urn:3gpp:ns:mcpttGKTP:1.0 |
| TS24484-ue-init-config | TS 24.484 [14] clause 7.2.2.3 | urn:3gpp:mcptt:mcpttUEinitConfig:1.0 |
| TS24484- mcptt-ue-config | TS 24.484 [14] clause 8.2.2.3 | urn:3gpp:mcptt:mcpttUEConfig:1.0 |

| | | |
|---|---|---|
| TS24484-mcvideo-ue-config | TS 24.484 [14] clause 9.2.2.3 | urn:3gpp:mcvideo:mcvideoUEConfig:1.0 |
| TS24484-mcdata-ue-config | TS 24.484 [14] clause 10.2.2.3 | urn:3gpp:mcdata:mcdataUEConfig:1.0 |
| TS24484-mcptt-user-profile | TS 24.484 [14] clause 8.3.2.3 | urn:3gpp:mcptt:user-profile:1.0 |
| TS24484-mcvideo-user-profile | TS 24.484 [14] clause 9.3.2.3 | urn:3gpp:ns:mcvideo:user-profile:1.0 |
| TS24484-mcdata-user-profile | TS 24.484 [14] clause 10.3.2.3 | urn:3gpp:ns:mcdata:user-profile:1.0 |
| TS24484-mcptt-service-config | TS 24.484 [14] clause 8.4.2.3 | urn:3gpp:ns:mcpttServiceConfig:1.0 |
| TS24484-mcvideo-service-config | TS 24.484 [14] clause 9.4.2.3 | urn:3gpp:ns:mcvideoServiceConfig:1.0 |
| TS24484-mcdata-service-config | TS 24.484 [14] clause 10.4.2.3 | urn:3gpp:ns:mcdataServiceConfig:1.0 |
| TS33180-mcsecKMSInterface | TS 33.180 [43] Annex D.3.5.1 | urn:3gpp:ns:mcsecKMSInterface:1.0 |
| TS33180-mcsecKMSKRR | TS 33.180 [43] Annex D.4.4 | urn:3gpp:ns:mcsecKMSKRR:1.0 |

NOTE 1: The schema does not define any target namespace but is referred to as "urn:ietf:params:xml:schema:patch-ops" e.g. in the original include statement in urn:ietf:params:xml:ns:xcap-diff.

NOTE 2: The XML schema in RFC 5261 [50] clause 8 is specified as DTD with ENTITY declarations being referred by the pattern of the schema. These references are resolved in RFC5261-patch-ops in order to get schema definitions which can be processed by TTCN-3 compilers.

NOTE 3: The schema specifies import of non-existing namespace urn:3gpp:ns:mcvideoGKTP:1.0.
Assumption: urn:3gpp:ns:mcpttGKTP:1.0 to be imported instead

NOTE 4: The schemata in TS 24.379 [9] F.4.2, TS 24.281 [55] F.4.2 and TS 24.282 [57] Annex D.1.2 are effectively the same apart from the namespace prefix ("mcpttaff" vs. "mcvideoaff" vs. "mcdataaff").
Assumption: The namespace prefix does not matter in XML documents, in general any prefix can be used.
⇒ TS24379_mcpttaff is used for MCPTT, MCVideo and MCData.

NOTE 5: The schemata in TS 24.379 [9] Table 7.4.1.2.2-2 and TS 24.281 [55] Table 7.4.1.2.2-2 are the same.
⇒ TS24379_poc-settings is used for MCPTT and MCVideo.

NOTE 6: The "XML schema for regroup using preconfigured group" in TS 24.379 [9], TS 24.281 [55] and TS 24.282 [57] have the same target namespace.

Editor's note: The XML schemas define identical types with the same name ("preconfig-group-Type", "groups-for-regroup-Type", "users-for-regroup-Type", "anyExtType") which is not allowed in one and the same namespace. As workaround these type definitions are removed from the XML schemas of TS 24.281 [55] and TS 24.282 [57].

NOTE 7: rest_netapi_nms-v1_0 uses the simple type xsd:dateTimeStamp which does not exist in XML schema 1.0 (but in XML schema 1.1) and therefore is replaced by xsd:dateTime.

In order to avoid ambiguities and unexpected side effects due to tool specific behaviour the schemaLocation in xs:import statements of XSD files are modified to use local references rather than references to internet locations.

The schemaLocations of imported namespaces are shown in table D.1-2. The schemaLocations of included namespaces are shown in table D.1-3.

**Table D.1-2: schemaLocation of imported namespaces**

| XML Schema | Imported name space | Original schemaLocation | Modified schemaLocation |
|---|---|---|---|
| RFC4354-poc-settings | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| RFC4661-SimpleFilter | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| IANA-resource-lists | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| poc_listService-v1_0 | urn:ietf:params:xml:ns:common-policy | http://www.iana.org/assignments/xml-registry/schema/common-policy.xsd | ../../Common/IMS_XSD/RFC4745-common-policy.xsd (NOTE 1) |
| poc_listService-v1_0 | urn:ietf:params:xml:ns:resource-lists | http://www.iana.org/assignments/xml-registry/schema/resource-lists.xsd | IANA-resource-lists.xsd (NOTE 2) |
| xdm_extensions-v1_0 | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/03/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| rest_netapi_common-v1_0 | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| rest_netapi_nms-v1_0 | urn:oma:xml:rest:netapi:common:1 | http://www.openmobilealliance.org/tech/profiles/rest_netapi_common-v1_0.xsd | rest_netapi_common-v1_0.xsd |
| rest_netapi_notificationchannel-v1_0 | urn:oma:xml:rest:netapi:common:1 | http://www.openmobilealliance.org/tech/profiles/rest_netapi_common-v1_0.xsd | rest_netapi_common-v1_0.xsd |
| xenc-schema | http://www.w3.org/2000/09/xmldsig# | xmldsig-core-schema.xsd | (no change needed) |
| TS24282_mcdatainfo.xsd | http://www.w3.org/2001/04/xmlenc# | http://www.w3.org/TR/xmlenc-core/xenc-schema.xsd | xenc-schema.xsd |
| TS24379_mcpttsigneddoc | http://www.w3.org/2000/09/xmldsig# | http://www.w3.org/TR/xmldsig-core/xmldsig-core-schema.xsd | xmldsig-core-schema.xsd |
| TS24484-ue-init-config | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| TS24484- mcptt-ue-config | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| TS24484-mcvideo-ue-config | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| TS24484-mcdata-ue-config | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| TS24484-mcptt-user-profile | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| TS24484-mcptt-user-profile | urn:ietf:params:xml:ns:common-policy | http://www.iana.org/assignments/xml-registry/schema/common-policy.xsd | ../../Common/IMS_XSD/RFC4745-common-policy.xsd (NOTE 1) |
| TS24484-mcvideo-user-profile | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| TS24484-mcvideo-user-profile | urn:ietf:params:xml:ns:common-policy | http://www.iana.org/assignments/xml-registry/schema/common-policy.xsd | ../../Common/IMS_XSD/RFC4745-common-policy.xsd (NOTE 1) |

| XML Schema | Imported name space | Original schemaLocation | Modified schemaLocation |
|---|---|---|---|
| TS24484-mcdata-user-profile | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| TS24484-mcdata-user-profile | urn:ietf:params:xml:ns:common-policy | http://www.iana.org/assignments/xml-registry/schema/common-policy.xsd | ../../Common/IMS_XSD/RFC4745-common-policy.xsd (NOTE 1) |
| TS24484-mcptt-service-config | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |
| TS24484-mcdata-service-config | http://www.w3.org/XML/1998/namespace | http://www.w3.org/2001/xml.xsd | ../../Common/IMS_XSD/xml.xsd |

NOTE 1: The namespace urn:ietf:params:xml:ns:common-policy is specified in RFC 4745 and at http://www.iana.org/assignments/xml-registry/schema/common-policy.xsd with difference in the minOccurs for identityType and validityType. Nevertheless for use in TTCN there shall be only one schema as part 9 of the TTCN-3 language [31] does not support different variants of one and the same namespace. In addition according to DIFF1 for RFC 4745 it seems that the RFC's variant is newer than the IANA's variant.

NOTE 2: The namespace urn:ietf:params:xml:ns:resource-lists is specified in RFC 4826 and at http://www.iana.org/assignments/xml-registry/schema/resource-lists.xsd with the difference of attribute anchor not being mandatory in the RFC.

**Table D.1-3: schemaLocation of included namespaces**

| XML Schema | Included schema | Original schemaLocation | Modified schemaLocation |
|---|---|---|---|
| RFC5874-xcap-diff | urn:ietf:params:xml:schema:patch-ops | urn:ietf:params:xml:schema:patch-ops | RFC5261-patch-ops.xsd |

# D.2 Common TTCN-3 Libraries

The same LibSip modules are used as according to TS 34.229-3 [28] annex G.0.2 and the same additional LibSip_MessageBodyTypes as according to TS 34.229-3 [28] annex G.1 and G.2

# Annex E (normative): TTCN-3 Definitions for the MCX IPCAN test model

## E.0 Introduction

The present Annex E specifies the TTCN-3 type definitions used at the system interface to configure and control the SS. The type definitions are specific for the MCX IPCAN test model.

In case of discrepancy between the content of the present Annex E and the equivalent TTCN-3 definitions / semantic requirements found in the TTCN modules provided as attachments to the present specification, the latter shall take precedence.

NOTE: This annex is automatically generated from the TTCN-3 modules provided as attachment to the present specification and containing the listed TTCN-3 type definitions.

## E.1 MCX_IPCAN_ASP_TypeDefs

**MCX_IPCAN_ASP_TypeDefs: Basic Type Definitions**

| TTCN-3 Basic Types | | |
|---|---|---|
| **MCX_IPCAN_AbsoluteCellPower_Type** | integer (-150..0) | absolute cell power (dBm); same as AbsoluteCellPower_Type and NR_AbsoluteCellPower_Type |
| **MCX_IPCAN_DeactivatedEdicatedEpsBearer_Type** | MCX_IPCAN_EpsBearerIdList_Type | Deactivate dedicated EPS bearer(s) for the MCX PDN<br>CellId : cell id of the cell<br>Reference implementation:<br>f_EUTRA_DeactivateEPS_BearerContext |
| **MCX_IPCAN_RemoveQosFlow_Type** | MCX_IPCAN_QosFlowList_Type | Remove QoS flow(s) from MCX PDU session<br>CellId : cell id of the cell<br>Reference implementation:<br>f_MCX_NR5GC_IPCAN_RemoveQosFlow |

**MCX_IPCAN_EpsBearerIdList_Type**

| TTCN-3 Record of Type | |
|---|---|
| **Name** | **MCX_IPCAN_EpsBearerIdList_Type** |
| **Comment** | |
| record of MCX_IPCAN_EpsBearerId_Type | |

**MCX_IPCAN_QosFlowList_Type**

| TTCN-3 Record of Type | |
|---|---|
| **Name** | **MCX_IPCAN_QosFlowList_Type** |
| **Comment** | |
| record of QosFlow_Identification_Type | |

**MCX_IPCAN_PacketFilterDirection_Type**

| TTCN-3 Enumerated Type | |
|---|---|
| **Name** | **MCX_IPCAN_PacketFilterDirection_Type** |
| **Comment** | packet filter direction according to 24.008 Table 10.5.162 |
| downlinkOnly | |
| uplinkOnly | |
| bidirectional | |

### MCX_IPCAN_PacketFilterComponentPortRange_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_PacketFilterComponentPortRange_Type** | | |
| **Comment** | | | |
| Low | integer | | |
| High | integer | | |

### MCX_IPCAN_PacketFilterComponent_Type

| TTCN-3 Union Type | | |
|---|---|---|
| **Name** | **MCX_IPCAN_PacketFilterComponent_Type** | |
| **Comment** | 24.008 Table 10.5.162 | |
| Protocol | integer | Protocol identifier/Next header type: 17 for UDP |
| SingleRemotePort | integer | |
| RemotePortRange | MCX_IPCAN_PacketFilterComponentPortRange_Type | |

### MCX_IPCAN_PacketFilterComponentList_Type

| TTCN-3 Record of Type | |
|---|---|
| **Name** | **MCX_IPCAN_PacketFilterComponentList_Type** |
| **Comment** | |
| record of MCX_IPCAN_PacketFilterComponent_Type | |

### MCX_IPCAN_PacketFilter_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_PacketFilter_Type** | | |
| **Comment** | | | |
| Direction | MCX_IPCAN_PacketFilterDirection_Type | | |
| ComponentList | MCX_IPCAN_PacketFilterComponentList_Type | | |

### MCX_IPCAN_PacketFilterList_Type

| TTCN-3 Record of Type | |
|---|---|
| **Name** | **MCX_IPCAN_PacketFilterList_Type** |
| **Comment** | |
| record of MCX_IPCAN_PacketFilter_Type | |

### MCX_IPCAN_DedicatedEpsBearerInfo_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_DedicatedEpsBearerInfo_Type** | | |
| **Comment** | Id and packet filter for dedicated EPS bearer (EUTRA) | | |
| EpsBearerId | MCX_IPCAN_EpsBearerId_Type | | EPS bearer id of the dedicated EPS bearer |
| PacketFilterList | MCX_IPCAN_PacketFilterList_Type | | |

### MCX_IPCAN_QosFlowInfo_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_QosFlowInfo_Type** | | |
| **Comment** | identification (PDU session id and QFI) and packet filter for dedicated Qos flow (NR5GC) | | |
| QosFlow | QosFlow_Identification_Type | | Qos flow identification |
| PacketFilterList | MCX_IPCAN_PacketFilterList_Type | | |

### MCX_IPCAN_IPv4IPv6Address_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_IPv4IPv6Address_Type** | | |
| **Comment** | | | |
| IPv4 | charstring | | |
| IPv6 | charstring | | |

### MCX_IPCAN_PDN_Info_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_PDN_Info_Type** | | |
| **Comment** | | | |
| PdnType | MCX_Registration_PDN_Type | | MCX, IMS or INTERNET |
| EpsBearerId | MCX_IPCAN_EpsBearerId_Type | | EPS bearer id of the PDN's default EPS bearer |
| IPAddressNW | MCX_IPCAN_IPv4IPv6Address_Type | | |
| IPAddressUE | MCX_IPCAN_IPv4IPv6Address_Type | | |

### MCX_IPCAN_PDN_InfoList_Type

| TTCN-3 Record of Type | |
|---|---|
| **Name** | **MCX_IPCAN_PDN_InfoList_Type** |
| **Comment** | list of PDNs to be connected to during Idle Update procedure |
| record length (1..3) of MCX_IPCAN_PDN_Info_Type | |

### MCX_IPCAN_PduSession_Info_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_PduSession_Info_Type** | | |
| **Comment** | | | |
| PdnType | MCX_Registration_PDN_Type | | MCX, IMS or INTERNET |
| APN | charstring | | |
| IPAddressNW | MCX_IPCAN_IPv4IPv6Address_Type | | |
| IPAddressUE | MCX_IPCAN_IPv4IPv6Address_Type | | |

### MCX_IPCAN_PduSession_InfoList_Type

| TTCN-3 Record of Type | |
|---|---|
| **Name** | **MCX_IPCAN_PduSession_InfoList_Type** |
| **Comment** | list of PDU sessions to be connected to during Idle Update procedure |
| record length (0..3) of MCX_IPCAN_PduSession_Info_Type | |

### MCX_IPCAN_ASP_CommonReqPart_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_ASP_CommonReqPart_Type** | | |
| **Comment** | Common Part of system requests | | |
| CellId | AnyCellId_Type | opt | not present in case of InitRequest and ReleaseRequest, mandatory otherwise |
| CnfFlag | boolean | | true: SS shall send a confirmation when the request has been executed<br>false: SS shall not send a confirmation for this request |

### MCX_IPCAN_ASP_CommonIndAckPart_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_ASP_CommonIndAckPart_Type** | | |
| **Comment** | Common Part of system indications and acknowledgements | | |
| CellId | AnyCellId_Type | | |

### MCX_IPCAN_Init_Type

| TTCN-3 Record Type | |
|---|---|
| **Name** | **MCX_IPCAN_Init_Type** |
| **Comment** | Initialisation of the system simulator.<br>CellId : not present<br>Reference implementation 4G: f_EUTRA_Init with p_SysinfoCombination:=c1, p_HandleULDataUM:=HANDLE_UM_DATA, p_UseBigGrants:=USE_BIG_GRANT<br>Reference implementation 5G: f_NR5GC_Init |

### MCX_IPCAN_CellConfig_Type

| TTCN-3 Record Type | |
|---|---|
| **Name** | **MCX_IPCAN_CellConfig_Type** |
| **Comment** | Create a cell and send out system information.<br>CellId : cell id of the cell to be created<br>Reference implementation 4G: f_EUTRA_CellConfig_Def<br>Reference implementation 5G: f_NR_CellConfig_Def |

### MCX_IPCAN_CellPower_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_CellPower_Type** | | |
| **Comment** | Set the cell power of a cell.<br>CellId : cell id of the cell<br>Reference implementation 4G: f_EUTRA_SetCellPower<br>Reference implementation 5G: f_NR_SetCellPower | | |
| AbsoluteCellPower | MCX_IPCAN_AbsoluteCellPower_Type | | |

### MCX_IPCAN_InitialRegistration4G_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_InitialRegistration4G_Type** | | |
| **Comment** | Perform initial registration procedure according to 37.579-1 clause 5.4.1A and Table 5.4.2.2.3-1 steps 2 - 16A to bring the UE into state 3 according to Figure 4.5.1-1 in 36.508.<br>As described in clause 5.4.1A of 37.579-1 after switch on the UE may register to an internet PDN, an IMS PDN and the MCX PDN.<br>NOTE: The UE does not need to register to MCX during initial registration after switch-on.<br>CellId : cell id of the cell<br>Reference implementation: f_MCX_EUTRA_IPCAN_IdleUpdated | | |
| PdnList | MCX_IPCAN_PDN_InfoList_Type | | list of PDNs the UE is expected to connect to during Idle Update procedure; the list may or may not contain MCX |
| DefaultAPN | octetstring | | APN to be used for the PDN (if any) for which the UE does not provide an APN; APN encoded according to 23.003 clause 9.1 and RFC 1035 clause 4.1.2 |

### MCX_IPCAN_InitialRegistration5G_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_InitialRegistration5G_Type** | | |
| **Comment** | Perform initial registration procedure according to 37.579-1 clause 5.4.1B and Table 5.4.2.3.3-1 steps 1 - 15 to bring the UE into state 1N-A according to 38.508-1 clause 4.4A.<br>As described in clause 5.4.1B of 37.579-1 after switch on the UE may register to an internet PDN, an IMS PDN and the MCX PDN.<br>NOTE: The UE does not need to register to MCX during initial registration after switch-on.<br>CellId : cell id of the cell<br>Reference implementation: f_MCX_NR5GC_IPCAN_InitialRegistration (f_NR5GC_Registration, a_NR5GC_AdditionalPDUSessionEstablishmentReq, f_MCX_NR5GC_IPCAN_AddPduSession) | | |
| PduSessionList | MCX_IPCAN_PduSession_InfoList_Type | | list of PDU sessions the UE is expected to connect to during Idle Update procedure; the list may or may not contain MCX and it may be empty |

### MCX_IPCAN_ExplicitMCXRegistration4G_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_ExplicitMCXRegistration4G_Type** | | |
| **Comment** | Procedure used, if the UE is configured not to connect automatically to the MCX PDN after switch-on: In this case after initial registration the UE gets triggered from TTCN by an MMI command to start MCX registration and this procedure is started to perform steps 19 - 26 of 37.579-1 Table 5.4.2.2.3-1.<br>CellId : cell id of the cell<br>Reference implementation: f_MCX_EUTRA_IPCAN_ConnectToAdditionalPDN | | |
| PdnInfo | MCX_IPCAN_PDN_Info_Type | | |

### MCX_IPCAN_ExplicitMCXRegistration5G_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_ExplicitMCXRegistration5G_Type** | | |
| **Comment** | Procedure used, if the UE is configured not to connect automatically to the MCX PDN after switch-on: In this case after initial registration the UE gets triggered from TTCN by an MMI command to start MCX registration and this procedure is started to perform steps 1 - 9 of 37.579-1 Table 5.4.2.3.3-2.<br>CellId : cell id of the cell<br>Reference implementation: f_MCX_NR5GC_IPCAN_AddPduSession | | |
| PduSessionInfo | MCX_IPCAN_PduSession_Info_Type | | |

### MCX_IPCAN_RRCConnectionSetup4G_MO_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_RRCConnectionSetup4G_MO_Type** | | |
| **Comment** | MO RRC connection establishment according to 37.579-1 table 5.4.3.2.3-1<br>CellId : cell id of the cell<br>Reference implementation: f_MCX_EUTRA_IPCAN_RRCConnectionEstablishment_MO | | |
| DedicatedEpsBearers | MCX_IPCAN_EpsBearerIdList_Type | | In case of a pre-established session: list of dedicated EPS bearers for which the DRBs need to be established; otherwise {} |

### MCX_IPCAN_RRCConnectionSetup5G_MO_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_RRCConnectionSetup5G_MO_Type** | | |
| **Comment** | MO RRC establishment according to 37.579-1 table 5.4.3.3.3-1<br>CellId : cell id of the cell<br>Reference implementation: f_MCX_NR5GC_IPCAN_RRCEstablishment_MO | | |
| QosFlows | MCX_IPCAN_QosFlowList_Type | | In case of a pre-established session: list of QoS flows for which the DRBs need to be established; otherwise {} |

### MCX_IPCAN_RRCConnectionSetup4G_MT_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_RRCConnectionSetup4G_MT_Type** | | |
| **Comment** | MT RRC connection establishment according to 37.579-1 table 5.4.4.2.3-1<br>CellId : cell id of the cell<br>Reference implementation: f_MCX_EUTRA_IPCAN_RRCConnectionEstablishment_MT | | |
| DedicatedEpsBearers | MCX_IPCAN_EpsBearerIdList_Type | | In case of a pre-established session: list of dedicated EPS bearers for which the DRBs need to be established; otherwise {} |

### MCX_IPCAN_RRCConnectionSetup5G_MT_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_RRCConnectionSetup5G_MT_Type** | | |
| **Comment** | MT RRC establishment according to 37.579-1 table 5.4.4.3.3-1<br>CellId : cell id of the cell<br>Reference implementation: f_MCX_NR5GC_IPCAN_RRCEstablishment_MT | | |
| QosFlows | MCX_IPCAN_QosFlowList_Type | | In case of a pre-established session: list of QoS flows for which the DRBs need to be established; otherwise {} |

### MCX_IPCAN_ActivateDedicatedEpsBearer_Type

| TTCN-3 Record of Type | |
|---|---|
| **Name** | **MCX_IPCAN_ActivateDedicatedEpsBearer_Type** |
| **Comment** | Establish dedicated EPS bearer(s) for the MCX PDN<br>CellId : cell id of the cell<br>Reference implementation: f_MCX_EUTRA_IPCAN_ActivateDedicatedEpsBearer |
| record of MCX_IPCAN_DedicatedEpsBearerInfo_Type | |

### MCX_IPCAN_AddQosFlow_Type

| TTCN-3 Record of Type | |
|---|---|
| **Name** | **MCX_IPCAN_AddQosFlow_Type** |
| **Comment** | Add QoS flow(s) to MCX PDU session<br>CellId : cell id of the cell<br>Reference implementation: f_MCX_NR5GC_IPCAN_AddQosFlow |
| record of MCX_IPCAN_QosFlowInfo_Type | |

### MCX_IPCAN_RRCConnectionRelease_Type

| TTCN-3 Record Type | |
|---|---|
| **Name** | **MCX_IPCAN_RRCConnectionRelease_Type** |
| **Comment** | Relase RRC connection<br>CellId : cell id of the cell<br>Reference implementation 4G: f_EUTRA_RRC_ConnectionRelease<br>Reference implementation 5G: f_NR_RRCRelease |

### MCX_IPCAN_Release_Type

| TTCN-3 Record Type | |
|---|---|
| **Name** | **MCX_IPCAN_Release_Type** |
| **Comment** | Release all configuration at the system simulator;<br>CellId : not present<br>Reference implementation 4G: f_EUTRA_Postamble with p_E_State:=E1_IDLE<br>Reference implementation 5G: f_NR_Postamble with p_UE_State:=STATE_IDLE_1A |

### MCX_IPCAN_SystemRequest_Type

| TTCN-3 Union Type | | |
|---|---|---|
| **Name** | **MCX_IPCAN_SystemRequest_Type** | |
| **Comment** | | |
| Init | MCX_IPCAN_Init_Type | |
| CellConfig | MCX_IPCAN_CellConfig_Type | |
| CellPower | MCX_IPCAN_CellPower_Type | |
| InitialRegistration4G | MCX_IPCAN_InitialRegistration4G_Type | |
| InitialRegistration5G | MCX_IPCAN_InitialRegistration5G_Type | |
| ExplicitMCXRegistration4G | MCX_IPCAN_ExplicitMCXRegistration4G_Type | |
| ExplicitMCXRegistration5G | MCX_IPCAN_ExplicitMCXRegistration5G_Type | |
| RRCConnectionSetup4G_MO | MCX_IPCAN_RRCConnectionSetup4G_MO_Type | |
| RRCConnectionSetup5G_MO | MCX_IPCAN_RRCConnectionSetup5G_MO_Type | |
| RRCConnectionSetup4G_MT | MCX_IPCAN_RRCConnectionSetup4G_MT_Type | |
| RRCConnectionSetup5G_MT | MCX_IPCAN_RRCConnectionSetup5G_MT_Type | |
| ActivateDedicatedEpsBearer | MCX_IPCAN_ActivateDedicatedEpsBearer_Type | |
| AddQosFlow | MCX_IPCAN_AddQosFlow_Type | |
| DeactivateDedicatedEpsBearer | MCX_IPCAN_DeactivateDedicatedEpsBearer_Type | |
| RemoveQosFlow | MCX_IPCAN_RemoveQosFlow_Type | |
| RRCConnectionRelease | MCX_IPCAN_RRCConnectionRelease_Type | |
| Release | MCX_IPCAN_Release_Type | |

### MCX_IPCAN_SYSTEM_CTRL_REQ

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_SYSTEM_CTRL_REQ** | | |
| **Comment** | | | |
| Common | MCX_IPCAN_ASP_CommonReqPart_Type | | |
| Request | MCX_IPCAN_SystemRequest_Type | | |

### MCX_IPCAN_InitConfirmation_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_InitConfirmation_Type** | | |
| **Comment** | | | |
| RanType | IPCAN_RAN_Type | | EUTRA_FDD or EUTRA_TDD for EUTRA, NR for NR5GC |

### MCX_IPCAN_SystemConfirmation_Type

| TTCN-3 Union Type | | |
|---|---|---|
| **Name** | **MCX_IPCAN_SystemConfirmation_Type** | |
| **Comment** | | |
| Init | MCX_IPCAN_InitConfirmation_Type | Confirmation for Init request: carries the RAN Type |
| Other | Null_Type | Confirmation for any other request |
| Error | charstring | Error indication when the corresponding request causes error; the charstring may be used to provide information about the error |

### MCX_IPCAN_SYSTEM_CTRL_CNF

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_SYSTEM_CTRL_CNF** | | |
| **Comment** | | | |
| Confirmation | MCX_IPCAN_SystemConfirmation_Type | | |

### MCX_IPCAN_PduSessionEstReqInd_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_PduSessionEstReqInd_Type** | | |
| **Comment** | | | |
| PduSessionId | integer | | |

### MCX_IPCAN_SystemIndication_Type

| TTCN-3 Union Type | | |
|---|---|---|
| **Name** | **MCX_IPCAN_SystemIndication_Type** | |
| **Comment** | | |
| PduSessionEstReqInd | MCX_IPCAN_PduSessionEstReqInd_Type | |

### MCX_IPCAN_SYSTEM_IND

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_SYSTEM_IND** | | |
| **Comment** | | | |
| Common | MCX_IPCAN_ASP_CommonIndAckPart_Type | | |
| Indication | MCX_IPCAN_SystemIndication_Type | | |

### MCX_IPCAN_PduSessionEstReqAck_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_PduSessionEstReqAck_Type** | | |
| **Comment** | | | |
| QFI | integer | | |

### MCX_IPCAN_SystemAcknowledgement_Type

| TTCN-3 Union Type | | |
|---|---|---|
| **Name** | **MCX_IPCAN_SystemAcknowledgement_Type** | |
| **Comment** | | |
| PduSessionEstReqAck | MCX_IPCAN_PduSessionEstReqAck_Type | |

### MCX_IPCAN_SYSTEM_ACK

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **MCX_IPCAN_SYSTEM_ACK** | | |
| **Comment** | | | |
| Common | MCX_IPCAN_ASP_CommonIndAckPart_Type | | |
| Acknowledgement | MCX_IPCAN_SystemAcknowledgement_Type | | |

### MCX_IPCAN_SYSTEM_CTRL_PORT

| TTCN-3 Port Type | | |
|---|---|---|
| **Name** | **MCX_IPCAN_SYSTEM_CTRL_PORT** | |
| **Comment** | | |
| out | MCX_IPCAN_SYSTEM_CTRL_REQ | |
| in | MCX_IPCAN_SYSTEM_CTRL_CNF | |

### MCX_IPCAN_SYSTEM_IND_PORT

| TTCN-3 Port Type | | |
|---|---|---|
| **Name** | **MCX_IPCAN_SYSTEM_IND_PORT** | |
| **Comment** | | |
| out | MCX_IPCAN_SYSTEM_ACK | |
| in | MCX_IPCAN_SYSTEM_IND | |

# E.2 MCX_CommonIPCAN

**MCX_CommonIPCAN: Basic Type Definitions**

| TTCN-3 Basic Types | | |
|---|---|---|
| **MCX_IPCAN_EpsBearerId_Type** | integer (5..13) | EPS bearer id: the associated DRB Id shall always be EpsBearerId - 4 |

**MCX_Registration_PDN_Type**

| TTCN-3 Enumerated Type | |
|---|---|
| **Name** | **MCX_Registration_PDN_Type** |
| **Comment** | Type definition for PIXITs px_MCX_InitialRegistration_TypeOf1stPDN, px_MCX_InitialRegistration_TypeOf2ndPDN, px_MCX_InitialRegistration_TypeOf3rdPDN; (see 37.579-1 clause 5.4.1A)<br>NOTE: 'ims', 'internet', 'mcx' and 'none' cannot be used as 'none' is a keyword; 'IMS', 'INTERNET', 'MCX' and 'NONE' cannot be used as 'IMS' is a port name of various PTCs |
| imsPDN | |
| internetPDN | |
| mcxPDN | |
| noPDN | |

# E.3 CommonDefs

**CommonDefs: Basic Type Definitions**

| TTCN-3 Basic Types | | |
|---|---|---|
| **Null_Type** | boolean (true) | dummy type for 'typeless' fields in unions |

**NR_CellId_Type**

| TTCN-3 Enumerated Type | |
|---|---|
| **Name** | **NR_CellId_Type** |
| **Comment** | |
| nr_Cell_NonSpecific | |
| nr_Cell1 | |
| nr_Cell2 | |
| nr_Cell3 | |
| nr_Cell4 | |
| nr_Cell6 | |
| nr_Cell10 | |
| nr_Cell11 | |
| nr_Cell12 | |
| nr_Cell13 | |
| nr_Cell14 | |
| nr_Cell23 | |
| nr_Cell28 | |
| nr_Cell29 | |
| nr_Cell30 | |
| nr_Cell31 | |

### EUTRA_CellId_Type

| TTCN-3 Enumerated Type | |
|---|---|
| **Name** | **EUTRA_CellId_Type** |
| **Comment** | |
| eutra_Cell_NonSpecific | |
| eutra_Cell1 | |
| eutra_Cell2 | |
| eutra_Cell3 | |
| eutra_Cell4 | |
| eutra_Cell6 | |
| eutra_Cell10 | |
| eutra_Cell11 | |
| eutra_Cell12 | |
| eutra_Cell13 | |
| eutra_Cell14 | |
| eutra_Cell23 | |
| eutra_Cell28 | |
| eutra_Cell29 | |
| eutra_Cell30 | |
| eutra_Cell31 | |
| eutra_CellA | |
| eutra_CellB | |
| eutra_CellC | |
| eutra_CellD | |
| eutra_CellE | |
| eutra_CellG | |
| eutra_CellH | |
| eutra_CellI | |
| eutra_CellJ | |
| eutra_CellK | |
| eutra_CellL | |
| eutra_CellM | |

### AnyCellId_Type

| TTCN-3 Union Type | | |
|---|---|---|
| **Name** | **AnyCellId_Type** | |
| **Comment** | | |
| Eutra | EUTRA_CellId_Type | |
| Nr | NR_CellId_Type | |

### QosFlow_Identification_Type

| TTCN-3 Record Type | | | |
|---|---|---|---|
| **Name** | **QosFlow_Identification_Type** | | |
| **Comment** | | | |
| PDU_SessionId | integer | | TS 24.007 clause 11.2.3.1b |
| QFI | integer | | TS 24.501 Table 11.2.3.1c.1 |

**IPCAN_RAN_Type**

| TTCN-3 Enumerated Type | |
|---|---|
| **Name** | **IPCAN_RAN_Type** |
| **Comment** | radio access network technology used by the IPCAN PTC |
| GERAN | |
| UTRAN_FDD | |
| UTRAN_TDD | |
| EUTRA_FDD | |
| EUTRA_TDD | |
| WLAN | |
| NR | |

## E.4 References to TTCN-3

| References to TTCN-3 | | |
|---|---|---|
| **MCX_IPCAN_ASP_TypeDefs** | IPCAN_IpcanModel/MCX_IPCAN_ASP_TypeDefs.ttcn | Rev 41472 |
| **MCX_CommonIPCAN** | IPCAN_Common/MCX_CommonIPCAN.ttcn | Rev 41607 |
| **CommonDefs** | Common/CommonDefs.ttcn | Rev 41588 |

# Annex F (informative): Procedures of the MCX IPCAN test model

## F.1 Introduction

This annex describes the major procedures for the SS based E-UTRA/EPC or NR/5GC implementation of the MCX IPCAN test model. The procedures are described in tables showing the primitives exchanged between TTCN and the system simulator at the IPCAN model specific ports SYS_REQ and SYS_IND as shown in Figure 5.2.1-1 and the related actions.

In general, the procedures correspond to the procedures specified in TS 37.579-1 [2] clause 5.4.

## F.2 Procedures

### F.2.1 Initialisation

This procedure is used to initialise the system simulator and to retrieve the RAN type (EUTRA_FDD, EUTRA_TDD, NR) to be used.

**Table F.2.1-1: Initialisation**

| Step | Action/Step | System Primitives at SYS_REQ port | Direction TTCN-SS | Relevant parameters / Comments |
|---|---|---|---|---|
| 1 | Common initialisation | Init (NOTE 1) | --> | - |
| 2 | Confirmation that initialisation is completed | Confirmation (Init) (NOTE 2) | <-- | RAN type |
| 3 | Cell configuration | CellConfig (NOTE 1) | --> | cell id |
| 4 | Confirmation that cell configuration is completed | Confirmation (Other) (NOTE 2) | <-- | - |
| NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type).<br>NOTE 2: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type). | | | | |

### F.2.2 Initial Registration

This procedure is used for initial registration and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.2.

**Table F.2.2-1: Initial Registration**

| Step | Action/Step | System Primitives at SYS_REQ/IND port | Direction TTCN-SS | Relevant parameters / Comments |
|---|---|---|---|---|
| 1 | Configuration of the cell power | CellPower (NOTE 1) | --> | cell id, absolute cell power |
| - | EXCEPTION: IF the RAN type is EUTRA THEN step 2a1 is perfomed | - | - | - |
| 2a1 | Start of IP handling (Address, IMS, HTTP servers) for MCX | - | - | cell id, MCX default EPS bearer id |
| 3 | Switch on the UE | - | - | - |
| 4 | Start of initial registration | EUTRA: InitialRegistration4G (NOTE 1) | --> | list of PDN infos: PDN type, EPS bearer id, IP addresses |
| | | NR5GC: InitialRegistration5G (NOTE 1) | --> | list of PDU session infos: PDN type, APN, IP addresses |
| - | EXCEPTION: Depending on the UE configuration, IF the UE does not register to MCX automatically at initial registration THEN steps 5a1-5a5 are performed | - | - | - |
| 5a1 | Confirmation that the initial registration is completed | Confirmation (Other) (NOTE 2) | <-- | (NOTE 5) |
| 5a2 | Request of RRC connection release | RRCConnectionRelease (NOTE 1) | --> | - |
| 5a3 | Confirmation for RRC connection release | Confirmation (Other) (NOTE 2) | <-- | |
| 5a4 | Make the UE (MCX client) request service authorisation/configuration | - | - | - |
| 5a5 | Start of explicit MCX registration | EUTRA: ExplicitMCXRegistration4G (NOTE 1) | --> | MCX PDN info: PDN type, EPS bearer id, IP addresses |
| | | NR5GC: ExplicitMCXRegistration5G (NOTE 1) | --> | MCX PDU session info: PDN type, APN, IP addresses |
| - | EXCEPTION: IF the RAN type is NR/5GC THEN Steps 6a1-6a3 are performed (NOTE 6) | - | - | - |
| 6a1 | Indication of PDU SESSION ESTABLISHMENT REQUEST | PduSessionEstReqInd (NOTE 3) | <-- | PDU session id |
| 6a2 | Start of IP handling (Address, IMS, HTTP servers) for MCX | - | - | cell id, MCX default QoS flow (PDU session id, QFI) |
| 6a3 | Acknowledgement of PDU session id and assignment of the QFI to be used for MCX default QoS flow | PduSessionEstReqAck (NOTE 4) | --> | QFI |
| 7 | HTTP and SIP signalling according to TS 37.579-1 [2] Table 5.3.2.2.1-1 | - | - | - |
| 8 | Request of RRC connection release | RRCConnectionRelease (NOTE 1) | --> | - |
| 9 | Confirmation for RRC connection release | Confirmation (Other) (NOTE 2) | <-- | - |

NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type).
NOTE 2: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type).
NOTE 3: MCX_IPCAN_SYSTEM_IND with Indication as specified in annex E (MCX_IPCAN_SystemIndication_Type).
NOTE 4: MCX_IPCAN_SYSTEM_ACK with Acknowledgement as specified in annex E (MCX_IPCAN_SystemAcknowledgement_Type)
NOTE 5: A confirmation is requested by the InitialRegistration at step 4 if the UE does not register to MCX automatically.
NOTE 6: Basically, the SS should indicate the PDU session id provided by the UE in the PDU SESSION ESTABLISHMENT REQUEST (step 6a1) and wait for the acknowledgement before sending the PDU SESSION ESTABLISHMENT ACCEPT. Nevertheless, it is up to SS implementation how to know the PDU session id, but the SS is expected to indicate the PDU session id early enough so that the HTTP server is in place when the initial HTTP request from the UE needs to be handled in TTCN.

## F.2.3 MO RRC connection setup

This procedure is used for MO RRC connection setup and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.3: Steps 2-8 of Table 5.4.3.2.3-1 for EUTRA and steps 1-7 of Table 5.4.3.3.3-1 for NR5GC, with m=1 at step 7 of Table 5.4.3.2.3-1 and at step 6 of Table 5.4.3.3.3-1 in case of a pre-established session and m=0 otherwise.

**Table F.2.3-1: MO RRC connection setup**

| Step | Action/Step | System Primitives at SYS_REQ port | Direction TTCN-SS | Relevant parameters / Comments |
|---|---|---|---|---|
| 1 | Request of MO RRC connection setup | EUTRA: RRCConnectionSetup4G_MO (NOTE 1) | --> | list of dedicated EPS bearer ids (NOTE 2) |
| | | NR5GC: RRCConnectionSetup5G_MO (NOTE 1) | --> | list of dedicated QoS flows (PDU session id, QFI) (NOTE 2) |
| NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type).<br>NOTE 2: In case of pre-established session, the RRC connection setup request provides a list of dedicated EPS bearer ids (EUTRA/EPC) or dedicated QoS flows (NR/5GC) with a single entry. When there is no pre-established session, the list is empty. | | | | |

## F.2.4 MT RRC connection setup

This procedure is used for MT RRC connection setup and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.4: Steps 1-8 of Table 5.4.4.2.3-1 for EUTRA and steps 1-8 of Table 5.4.4.3.3-1 for NR5GC, with m=1 at step 7 of Table 5.4.4.2.3-1 and at step 7 of Table 5.4.4.3.3-1 in case of a pre-established session and m=0 otherwise.

**Table F.2.4-1: MT RRC connection setup**

| Step | Action/Step | System Primitives at SYS_REQ port | Direction TTCN-SS | Relevant parameters / Comments |
|---|---|---|---|---|
| 1 | Request of MT RRC connection setup | EUTRA: RRCConnectionSetup4G_MT (NOTE 1) | --> | list of dedicated EPS bearer ids (NOTE 2) |
| | | NR5GC: RRCConnectionSetup5G_MT (NOTE 1) | --> | list of dedicated QoS flows (PDU session id, QFI) (NOTE 2) |
| 2 | Confirmation for RRC connection establishment | Confirmation (Other) (NOTE 3) | <-- | - |
| NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type).<br>NOTE 2: In case of pre-established session, the RRC connection setup request provides a list of dedicated EPS bearer ids (EUTRA/EPC) or dedicated QoS flows (NR/5GC) with a single entry. When there is no pre-established session, the list is empty.<br>NOTE 3: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type). | | | | |

## F.2.5 MO communication establishment

This procedure is used for MO communication establishment and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.3: Steps 2-16a3 of Table 5.4.3.2.3-1 for EUTRA and steps 1-8a3 of Table 5.4.3.3.3-1 for NR5GC, with m=0 at step 7 of Table 5.4.3.2.3-1 and at step 6 of Table 5.4.3.3.3-1.

**Table F.2.5-1: MO communication establishment**

| Step | Action/Step | System Primitives at SYS_REQ port | Direction TTCN-SS | Relevant parameters / Comments |
|---|---|---|---|---|
| 1 | MO RRC connection setup as specified in Table F.2.3-1 | - | - | - |
| 2 | SIP signalling: MO INVITE | - | - | - |
| - | EXCEPTION: In parallel to steps 3a1-5 the SIP signalling for completion of the MO call establishment happens (NOTE 3) | - | - | - |
| - | EXCEPTION: Steps 3a1-3b1 depend on the kind of MCX service. | - | - | - |
| 3a1 | IF the MCX service is MCPTT or MCVideo THEN RTP/RTCP loopback gets configured | | | cell id and DRB id (EUTRA) or QoS flow (PDU session id, QFI; NR5GC) |
| 3b1 | ELSE IF the MCX service is MCData THEN MSRP server gets started in TTCN | - | - | cell id and DRB id (EUTRA) or QoS flow (PDU session id, QFI; NR5GC) |
| 4 | Activate/Add dedicated resources | EUTRA: ActivateDedicatedEpsBearer (NOTE 1) | --> | dedicated EPS bearer id, packet filters |
| | | NR5GC: AddQosFlow (NOTE 1) | --> | dedicated QoS flow (PDU session id, QFI), packet filters |
| 5 | Confirmation for establishment of dedicated resources | Confirmation (Other) (NOTE 2) | <-- | - |
| NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type).<br>NOTE 2: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type).<br>NOTE 3: The UE is expected to send the SIP ACK not before dedicated resources are established. | | | | |

## F.2.6 MT communication establishment

This procedure is used for MT communication establishment and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.4: Steps 1-17a3 of Table 5.4.4.2.3-1 for EUTRA and steps 1-9a3 of Table 5.4.4.3.3-1 for NR5GC, with m=0 at step 7 of Table 5.4.4.2.3-1 and at step 7 of Table 5.4.4.3.3-1.

**Table F.2.6-1: MT communication establishment**

| Step | Action/Step | System Primitives at SYS_REQ port | Direction TTCN-SS | Relevant parameters / Comments |
|---|---|---|---|---|
| 1 | MT RRC connection setup as specified in Table F.2.4-1 | - | - | - |
| - | EXCEPTION: In parallel to steps 3a1-4 the SIP signalling for MT call establishment happens | - | - | - |
| - | EXCEPTION: Steps 3a1-3b1 depend on the kind of MCX service. | - | - | - |
| 3a1 | IF the MCX service is MCPTT or MCVideo THEN RTP/RTCP loopback gets configured | | | cell id and DRB id (EUTRA) or QoS flow (PDU session id, QFI; NR5GC) |
| 3b1 | ELSE IF the MCX service is MCData THEN MSRP server gets started in TTCN | - | - | cell id and DRB id (EUTRA) or QoS flow (PDU session id, QFI; NR5GC) |
| 4 | Activate/Add dedicated resources | EUTRA: ActivateDedicatedEpsBearer (NOTE 1) | --> | dedicated EPS bearer id, packet filters |
| | | NR5GC: AddQosFlow (NOTE 1) | --> | dedicated QoS flow (PDU session id, QFI), packet filters |
| 5 | Confirmation for establishment of dedicated resources | Confirmation (Other) (NOTE 2) | <-- | - |
| NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type).<br>NOTE 2: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type). | | | | |

## F.2.7 Connection release

This procedure is used for communication release and corresponds to the procedures of TS 37.579-1 [2] clause 5.4.14.

**Table F.2.7-1: Connection release**

| Step | Action/Step | System Primitives at SYS_REQ port | Direction TTCN-SS | Relevant parameters / Comments |
|---|---|---|---|---|
| - | EXCEPTION: IF there is a dedicated bearer or QoS flow AND no pre-established session is established OR a pre-established session shall be released THEN steps 1a1-1a2 are performed. | - | - | - |
| 1a1 | Deactivate/Remove dedicated resources | EUTRA: DeactivateDedicatedEpsBearer (NOTE 1) | --> | dedicated EPS bearer id |
| | | NR5GC: RemoveQosFlow (NOTE 1) | --> | dedicated QoS flow (PDU session id, QFI |
| 1a2 | Confirmation for removal of dedicated resources | Confirmation (Other) (NOTE 2) | <-- | - |
| 2 | Request of RRC connection release | RRCConnectionRelease (NOTE 1) | --> | - |
| 3 | Confirmation for RRC connection release | Confirmation (Other) (NOTE 2) | <-- | - |
| NOTE 1: MCX_IPCAN_SYSTEM_CTRL_REQ with Request as specified in annex E (MCX_IPCAN_SystemRequest_Type).<br>NOTE 2: MCX_IPCAN_SYSTEM_CTRL_CNF) with Confirmation as specified in annex E (MCX_IPCAN_SystemConfirmation_Type). | | | | |

# Annex G (informative): Change history

| Change history of TS 36.579-5 up to v17.7.0 | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **Meeting** | **TDoc** | **CR** | **Rev** | **Cat** | **Subject/Comment** | **New version** |
| 2017-02 | RAN5#74 | R5-171302 | - | - | - | Introduction of TS 36.579-5. | 0.0.1 |
| 2018-03 | RAN5#78 | R5-180687 | - | - | - | Implements changes agreed in<br>R5-180618 "MCPTT: Initial Test Model"<br>R5-180657 "Various updates to 36579-5" | 0.1.0 |
| 2018-03 | RAN#79 | RP-180130 | - | - | - | Draft version for information purposes to the RAN Plenary | 1.0.0 |
| 2018-05 | RAN5#79 | R5-182437 | - | - | - | Implements changes agreed in<br>R5-183163<br>R5-183164 | 2.0.0 |
| 2018-06 | RAN#80 | RP-180655 | - | - | - | put under revision control as v13.0.0 with small editorial changes | 13.0.0 |
| 2018-09 | RAN#81 | R5-184081 | 0001 | - | F | MCPTT: Test Model updates | 13.1.0 |
| 2018-12 | RAN#82 | R5-192380 | 0002 | 1 | F | Routine maintenance for TS 36.579-5 | 13.2.0 |
| 2019-06 | RAN#84 | R5-195221 | 0003 | 1 | F | Routine maintenance for TS 36.579-5 | 13.3.0 |
| 2019-06 | RAN#84 | R5-195222 | 0004 | 1 | F | Introduction of MCPTT test model over IP | 13.3.0 |
| 2019-12 | RAN#86 | R5-199050 | 0005 | 1 | F | Routine maintenance for TS 36.579-5 | 13.4.0 |
| 2020-03 | RAN#87 | R5-201152 | 0006 | 1 | F | Routine maintenance for TS 36.579-5 | 13.5.0 |
| 2020-06 | RAN#88 | R5-203077 | 0007 | 1 | F | Routine maintenance for TS 36.579-5 | 13.6.0 |
| 2020-09 | RAN#89 | R5-203554 | 0008 | - | F | Routine maintenance for TS 36.579-5 | 13.7.0 |
| 2020-09 | RAN#89 | R5-204377 | 0010 | - | F | Updates to introduce support for MCVideo and MCData | 14.0.0 |
| 2020-12 | RAN#90 | R5-206465 | 0011 | 1 | F | Routine maintenance for TS 36.579-5 | 14.1.0 |
| 2021-03 | RAN#91 | R5-210255 | 0012 | - | F | Routine maintenance for TS 36.579-5 | 14.2.0 |
| 2021-06 | RAN#92 | R5-213670 | 0017 | 1 | F | Routine maintenance for TS 36.579-5 | 14.3.0 |
| 2021-09 | RAN#93 | R5-216343 | 0019 | 1 | F | Routine maintenance for TS 36.579-5 | 14.4.0 |
| 2021-09 | RAN#93 | R5s210779 | 0018 | - | B | Addition of MCPTT test case 6.1.1.2 to the MCX ATS | 14.4.0 |
| 2021-09 | RAN#93 | R5s210897 | 0013 | 1 | B | Addition of MCPTT test case 5.1 to the MCX ATS | 14.4.0 |
| 2021-09 | RAN#93 | R5s210899 | 0014 | 1 | B | Addition of MCPTT test case 6.1.1.1 to the MCX ATS | 14.4.0 |
| 2021-09 | RAN#93 | R5s211178 | 0020 | - | F | Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A | 14.4.0 |
| 2021-12 | RAN#94 | R5-216753 | 0060 | - | F | Routine maintenance for TS 36.579-5 | 14.5.0 |
| 2021-12 | RAN#94 | R5s211359 | 0046 | - | B | Addition of MCPTT test case 6.2.1 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211361 | 0047 | - | B | Addition of MCPTT test case 6.2.2 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211363 | 0048 | - | B | Addition of MCPTT test case 6.2.3 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211365 | 0049 | - | B | Addition of MCPTT test case 6.2.4 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211367 | 0050 | - | B | Addition of MCPTT test case 6.2.5 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211369 | 0051 | - | B | Addition of MCPTT test case 6.2.6 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211371 | 0052 | - | B | Addition of MCPTT test case 6.2.7 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211373 | 0053 | - | B | Addition of MCPTT test case 6.2.8 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211377 | 0055 | - | B | Addition of MCPTT test case 6.2.15 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211381 | 0057 | - | B | Addition of MCPTT test case 6.2.17 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211463 | 0021 | 1 | B | Addition of MCPTT test case 5.3 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211469 | 0059 | - | F | Configuration of SRTP | 14.5.0 |
| 2021-12 | RAN#94 | R5s211617 | 0023 | 1 | B | Addition of MCPTT test case 6.1.1.4 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211618 | 0024 | 1 | B | Addition of MCPTT test case 6.1.1.8 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211619 | 0025 | 1 | B | Addition of MCPTT test case 6.1.1.9 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211620 | 0026 | 1 | B | Addition of MCPTT test case 6.1.1.11 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211621 | 0027 | 1 | B | Addition of MCPTT test case 6.1.1.12 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211628 | 0028 | 1 | B | Addition of MCPTT test case 6.1.1.13 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211629 | 0029 | 1 | B | Addition of MCPTT test case 6.1.1.14 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211630 | 0030 | 1 | B | Addition of MCPTT test case 6.1.2.7 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211631 | 0031 | 1 | B | Addition of MCPTT test case 6.1.2.8 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211632 | 0032 | 1 | B | Addition of MCPTT test case 6.1.2.9 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211633 | 0033 | 1 | B | Addition of MCPTT test case 6.1.2.10 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211634 | 0034 | 1 | B | Addition of MCPTT test case 6.1.2.11 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211635 | 0035 | 1 | B | Addition of MCPTT test case 6.1.2.12 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211688 | 0022 | 1 | B | Addition of MCPTT test case 6.1.1.3 to the MCX ATS | 14.5.0 |
| 2021-12 | RAN#94 | R5s211695 | 0061 | - | F | Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A | 14.5.0 |
| 2022-03 | RAN#95 | R5-220488 | 0070 | - | F | Routine maintenance for TS 36.579-5 | 14.6.0 |
| 2022-03 | RAN#95 | R5s220129 | 0062 | - | B | Addition of MCPTT test case 5.4 to the MCX ATS | 14.6.0 |
| 2022-03 | RAN#95 | R5s220131 | 0063 | - | B | Addition of MCPTT test case 6.1.1.5 to the MCX ATS | 14.6.0 |
| 2022-03 | RAN#95 | R5s220133 | 0064 | - | B | Addition of MCPTT test case 6.1.1.6 to the MCX ATS | 14.6.0 |
| 2022-03 | RAN#95 | R5s220135 | 0065 | - | B | Addition of MCPTT test case 6.1.1.10 to the MCX ATS | 14.6.0 |
| 2022-03 | RAN#95 | R5s220137 | 0066 | - | B | Addition of MCPTT test case 6.1.1.16 to the MCX ATS | 14.6.0 |
| 2022-03 | RAN#95 | R5s220139 | 0067 | - | B | Addition of MCPTT test case 6.1.1.18 to the MCX ATS | 14.6.0 |
| 2022-03 | RAN#95 | R5s220141 | 0068 | - | B | Addition of MCPTT test case 6.1.1.21 to the MCX ATS | 14.6.0 |
| 2022-03 | RAN#95 | R5s220143 | 0069 | - | B | Addition of MCPTT test case 6.2.16 to the MCX ATS | 14.6.0 |
| 2022-03 | RAN#95 | R5s220344 | 0071 | - | F | Correction of MCPTT ambient listening test cases | 14.6.0 |
| 2022-03 | RAN#95 | R5s220346 | 0072 | - | F | Update of function f_URL_Decoding | 14.6.0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2022-03 | RAN#95 | R5s220437 | 0073 | - | F | Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A | 14.6.0 |
| 2022-06 | RAN#96 | R5s220559 | 0074 | - | F | Correction of f_MCX_InviteMT_SendINVITE | 14.7.0 |
| 2022-09 | RAN#97 | R5-223960 | 0076 | - | F | Routine maintenance for TS 36.579-5 | 14.8.0 |
| 2022-09 | RAN#97 | R5s220869 | 0075 | - | F | MCX Implementation Issues | 14.8.0 |
| 2022-09 | RAN#97 | R5s220992 | 0077 | - | B | Addition of MCPTT test case 5.6 to the MCX ATS | 14.8.0 |
| 2022-09 | RAN#97 | R5s220994 | 0078 | - | B | Addition of MCPTT test case 6.1.1.17 to the MCX ATS | 14.8.0 |
| 2022-09 | RAN#97 | R5s220996 | 0079 | - | B | Addition of MCPTT test case 6.1.1.20 to the MCX ATS | 14.8.0 |
| 2022-09 | RAN#97 | R5s221072 | 0086 | - | F | Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A | 15.0.0 |
| 2022-12 | RAN#98 | R5s221283 | 0082 | 1 | B | Addition of MCData test case 5.1 to the MCX ATS | 15.1.0 |
| 2022-12 | RAN#98 | R5s221284 | 0083 | 1 | B | Addition of MCData test case 6.1.1 to the MCX ATS | 15.1.0 |
| 2022-12 | RAN#98 | R5s221285 | 0084 | 1 | B | Addition of MCData test case 6.1.3 to the MCX ATS | 15.1.0 |
| 2022-12 | RAN#98 | R5s221286 | 0085 | 1 | B | Addition of MCData test case 6.1.4 to the MCX ATS | 15.1.0 |
| 2022-12 | RAN#98 | R5s221361 | 0091 | - | F | Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A | 15.1.0 |
| 2022-12 | RAN#98 | - | - | - | - | upgrade to Rel16 with no change | 16.0.0 |
| 2022-12 | RAN#98 | R5s221241 | 0088 | - | F | Rel-17 Sep'22 baseline upgrade for MCX Test Suites | 17.0.0 |
| 2023-03 | RAN#99 | R5-230148 | 0092 | - | F | Routine maintenance for TS 36.579-5 | 17.1.0 |
| 2023-03 | RAN#99 | R5s230192 | 0090 | 1 | F | Correction of function fl_MCPTT_Info_AmbientListening_RX | 17.1.0 |
| 2023-06 | RAN#100 | R5-232221 | 0093 | - | F | Routine maintenance for TS 36.579-5 | 17.2.0 |
| 2023-06 | RAN#100 | R5-232604 | 0094 | - | F | Update to MCX PIXITs for MCData Functional Alias | 17.2.0 |
| 2023-09 | RAN#101 | R5-233868 | 0099 | - | F | Routine maintenance for TS 36.579-5 | 17.3.0 |
| 2023-09 | RAN#101 | R5s230578 | 0095 | 1 | B | Addition of MCPTT test case 5.8 to the MCX ATS | 17.3.0 |
| 2023-09 | RAN#101 | R5s230579 | 0096 | 1 | B | Addition of MCPTT test case 6.1.1.19 to the MCX ATS | 17.3.0 |
| 2023-09 | RAN#101 | R5s230580 | 0097 | 1 | B | Addition of MCPTT test case 6.1.2.13 to the MCX ATS | 17.3.0 |
| 2023-09 | RAN#101 | R5s230581 | 0098 | 1 | B | Addition of MCPTT test case 6.2.14 to the MCX ATS | 17.3.0 |
| 2023-09 | RAN#101 | R5s230672 | 0100 | - | F | Add new verified and e-mail agreed TTCN test cases in the TC lists in 36.579-5 (prose), Annex A | 17.3.0 |
| 2023-12 | RAN#102 | R5-237446 | 0102 | 1 | F | Routine maintenance for TS 36.579-5 | 17.4.0 |
| 2024-03 | RAN#103 | R5-241686 | 0105 | 1 | F | Addition of new PIXITs for REL-16 TCs | 17.5.0 |
| 2024-03 | RAN#103 | R5s240111 | 0103 | - | F | Correction of MMI in test cases 6.1.1.19 and 6.1.2.13 | 17.5.0 |
| 2024-03 | RAN#103 | R5s240148 | 0104 | - | F | Correction of f_MCX_IMS_ServiceAuthorization | 17.5.0 |
| 2024-03 | RAN#103 | R5s240241 | 0107 | - | F | Update of TTCN type definitions, Annex E | 17.5.0 |
| 2024-06 | RAN#104 | R5-242399 | 0111 | - | F | Routine maintenance for TS 36.579-5 | 17.6.0 |
| 2024-06 | RAN#104 | R5s240279 | 0108 | - | F | Corrections for MCX registration | 17.6.0 |
| 2024-06 | RAN#104 | R5s240332 | 0106 | 1 | F | Misc corrections for MCX | 17.6.0 |
| 2024-06 | RAN#104 | R5s240360 | 0112 | - | F | Correction CSK upload | 17.6.0 |
| 2024-06 | RAN#104 | R5s240420 | 0114 | - | F | Update of TTCN type definitions, Annex E | 17.6.0 |
| 2024-09 | RAN#105 | R5-245598 | 0120 | - | F | Routine maintenance for TS 36.579-5 | 17.7.0 |
| 2024-09 | RAN#105 | R5-245674 | 0119 | 1 | F | Addition of MCData Message Store Function terminology | 17.7.0 |
| 2024-09 | RAN#105 | R5s240492 | 0115 | 1 | F | Correction to the request URI template | 17.7.0 |

NOTE: The table above will not be further updated in the future. It shows all TS 36.579-5 CRs taken over into TS 37.579-5 v0.0.1.

| Change history of TS 37.579-5 | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **Meeting** | **TDoc** | **CR** | **Rev** | **Cat** | **Subject/Comment** | **New version** |
| 2024-11 | RAN5#105 | R5-246402 | - | - | - | TS 36.579-5 v17.7.0 content was transferred into this new TS 37.579-5 as by definition 36 series specifications cover LTE only aspects and multi-RAT aspects need to be covered in 37 series specifications.<br>The only changes, compared to TS 36.579-5 v17.7.0, are:<br>- "over LTE" has been removed from the TS title as beginning with Rel-17 the present document needs to cover also NR/5GC.<br>- text in Scope clause has been updated, not to limit the RAT to LTE.<br>- every instance of 36.579 has been replaced by 37.579, except in the Change history table of TS 36.579-5.<br>- every instance of 36.579-3 has been either deleted or voided.<br>- the CR history table of TS 36.579-5 was kept for easier reference of all changes included in TS 37.579-5, but a new Change history table was added for TS 37.579-5.<br>- the 3GPP TS-TR template version 1.18.1 has been used. | 0.0.1 |
| 2024-11 | RAN5#105 | R5-246420 | - | - | - | 36.579-5 CR#0123 has been implemented on top of 37.579-5-001 to produce 37.579-5-010. | 0.1.0 |
| 2024-12 | RAN#106 | RP-242531 | - | - | - | For one-step approval at RAN#106 | 1.0.0 |
| 2024-12 | RAN#106 | RP-243251 | - | - | - | The associated TTCN-3 code has been added as an attachment | 1.0.1 |
| 2024-12 | RAN#106 | - | - | - | - | raised to v17.0.0 with no change | 17.0.0 |
| 2024-12 | RAN#106 | - | - | - | - | raised to v18.0.0 with no change | 18.0.0 |
| 2025-03 | RAN#107 | R5-250360 | 0007 | - | F | Routine maintenance for TS 37.579-5 | 18.1.0 |
| 2025-03 | RAN#107 | R5-251407 | 0006 | 1 | F | Addition of initial Test Model for support of MCX Client over NR/5GC | 18.1.0 |
| 2025-03 | RAN#107 | R5s250034 | 0001 | - | F | Rel-18 Jun'24 baseline upgrade for MCX Test Suites | 18.1.0 |
| 2025-03 | RAN#107 | R5s250035 | 0002 | - | F | Correction of f_MCX_JSON_CreateTokens | 18.1.0 |
| 2025-03 | RAN#107 | R5s250036 | 0003 | - | F | Rel-18 Sep'24 baseline corrections for MCX Test Suites | 18.1.0 |
| 2025-03 | RAN#107 | R5s250037 | 0004 | - | B | Addition of MCPTT test case 5.1 to the MCX ATS with MCX-EUTRA model | 18.1.0 |
| 2025-03 | RAN#107 | R5s250038 | 0005 | - | F | Integrity protection added to NOTIFY for xcap-diff | 18.1.0 |
| 2025-03 | RAN#107 | R5s250078 | 0009 | - | F | Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.579-5 (prose), Annex A | 18.1.0 |
| 2025-06 | RAN#108 | R5-251937 | 0011 | - | F | NR/5GC enhancement of MCX IPCAN Test Model | 18.2.0 |
| 2025-06 | RAN#108 | R5-252226 | 0012 | - | F | Routine maintenance for TS 37.579-5 | 18.2.0 |
| 2025-06 | RAN#108 | R5-252297 | 0013 | - | F | Addition of Functional Alias PIXIT for MCVideo | 18.2.0 |
| 2025-06 | RAN#108 | R5s250133 | 0008 | 1 | B | Addition of MCPTT test case 6.1.1.16 to the MCX ATS with MCX-EUTRA model | 18.2.0 |
| 2025-06 | RAN#108 | R5s250157 | 0010 | - | F | Correction to function f_MCX_Registration_PDN_List_Match | 18.2.0 |
| 2025-06 | RAN#108 | R5s250219 | 0015 | - | F | Update of TTCN type definitions, Annex E | 18.2.0 |
| 2025-09 | RAN#109 | R5-254019 | 0017 | - | F | Routine maintenance for TS 37.579-5 | 18.3.0 |
| 2025-09 | RAN#109 | R5s250357 | 0018 | - | F | Update of TTCN type definitions, Annex E | 18.3.0 |
| 2025-12 | RAN#110 | R5-255799 | 0022 | - | F | Routine maintenance for TS 37.579-5 | 18.4.0 |
| 2025-12 | RAN#110 | R5-256721 | 0023 | 1 | F | Addition of MCData notification channel details | 18.4.0 |
| 2025-12 | RAN#110 | R5s250406 | 0016 | 1 | B | Addition of MCPTT test case 5.1 to the MCX ATS with MCX-NR5GC model | 18.4.0 |
| 2025-12 | RAN#110 | R5s250429 | 0019 | 1 | B | Addition of MCPTT test case 6.1.1.16 to the MCX ATS with MCX-NR5GC model | 18.4.0 |
| 2025-12 | RAN#110 | R5s250505 | 0024 | 1 | F | Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.579-5 (prose), Annex A | 18.4.0 |
| 2026-03 | RAN#111 | R5-260291 | 0027 | - | F | Routine maintenance for TS 37.579-5 | 18.5.0 |
| 2026-03 | RAN#111 | R5s260037 | 0025 | - | B | Addition of MCPTT test case 5.8 to the MCX ATS with MCX-EUTRA model | 18.5.0 |
| 2026-03 | RAN#111 | R5s260039 | 0026 | - | B | Addition of MCPTT test case 5.8 to the MCX ATS with MCX-NR5GC model | 18.5.0 |
| 2026-03 | RAN#111 | R5s260060 | 0020 | 1 | B | Addition of MCPTT test case 5.3 to the MCX ATS with MCX-EUTRA model | 18.5.0 |
| 2026-03 | RAN#111 | R5s260062 | 0021 | 1 | B | Addition of MCPTT test case 5.3 to the MCX ATS with MCX-NR5GC model | 18.5.0 |
| 2026-03 | RAN#111 | R5s260076 | 0028 | - | F | Add new verified and e-mail agreed TTCN test cases in the TC lists in 37.579-5 (prose), Annex A | 18.5.0 |

# History

| Version | Date | Status |
|---|---|---|
| V18.0.0 | February 2025 | Publication |
| V18.1.0 | April 2025 | Publication |
| V18.2.0 | July 2025 | Publication |
| V18.3.0 | October 2025 | Publication |
| V18.4.0 | February 2026 | Publication |
| V18.5.0 | April 2026 | Publication |